DRONE VIDEO ANALYTICS SYSTEM

Architecture, Deployment and Legal Compliance Document

Operational Intelligence Platform for Drone-Based Surveillance and Decision Support

Document Purpose

This document presents the complete architecture, operational workflow, deployment model, analytics capabilities, security controls, legal compliance framework, and rollout strategy for a Drone Video Analytics System proposed for operational use by the Kerala Police Department. The system is designed as a **decision-support platform**, assisting officers before, during, and after events, while ensuring strict adherence to privacy, evidentiary integrity, auditability, and statutory compliance.

Scope of the Document

This document covers:

- End-to-end system architecture from drone to dashboard
- Edge, ingest, GPU analytics, and storage layers
- Multi-drone scalability and deployment sizing
- Analytics capabilities and operational limitations
- Evidence handling, retention, and chain of custody
- Authentication, access control, and audit mechanisms
- Legal and regulatory compliance considerations• Phased rollout roadmap and operational readiness

Prepared For
Kerala Police Department

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Draft for Technical and Operational Review

Executive Summary

The increasing use of drones by law enforcement agencies has created an opportunity to significantly enhance situational awareness, public safety monitoring, and incident response. At the same time, it introduces operational, legal, and technical challenges related to real-time video processing, scalability, evidence integrity, and privacy compliance. This document presents a comprehensive Drone Video Analytics System designed to support Kerala Police operations through advanced analytics, structured workflows, and legally defensible evidence handling, while retaining full human control over decision-making.

The proposed system is conceived as an operational intelligence platform rather than an automated enforcement mechanism. Its primary objective is to assist officers by filtering, analysing, and presenting relevant information from drone video feeds in a timely and reliable manner. All analytics outputs are intended to support situational assessment and informed action by authorised personnel. At no point does the system replace officer judgment or automate coercive actions.

From a technical perspective, the architecture follows a layered design comprising the drone layer, operator laptop edge layer, central ingest and streaming layer, GPU-based analytics cluster, evidence storage systems, structured databases, and role-based dashboards. This separation of concerns ensures scalability, fault isolation, and clarity of responsibility. Edge processing on operator laptops performs buffering, change detection, and cognitive frame selection to reduce bandwidth usage and ensure that only meaningful data is transmitted for central analysis. Central analytics are performed on shared GPU resources, enabling efficient processing of multiple concurrent drone feeds without requiring a dedicated GPU per drone.

The system has been designed to operate in both pilot-scale and large-scale deployments. It supports multiple drones operating simultaneously from different locations, with batch-based GPU scheduling that optimises resource utilisation. As the number of drones increases, capacity can be expanded incrementally by adding ingest servers, GPU nodes, or storage resources, without requiring architectural redesign. This makes the system suitable for phased deployment, starting from limited pilots and progressing to district-level and state-level operations.

A core focus of the system is evidence integrity and legal admissibility. Evidence clips are generated automatically based on confirmed events, with contextual frames captured before and after the incident. Each evidence artifact is bound with timestamps, source identifiers, and integrity metadata at the time of creation. Evidence is stored in secure on-premise object storage with retention policies aligned to operational needs and legal requirements. All access to evidence and system functions is governed by strong authentication, role-based and attribute-based access control, and comprehensive audit logging. This ensures a continuous and verifiable chain of custody, suitable for judicial scrutiny under the Indian Evidence Act and related statutes.

The analytics capabilities demonstrated include person detection and tracking, face detection and image de-duplication, crowd detection and density estimation, zone-based analysis, intrusion and perimeter breach detection, accident and incident detection, abandoned or suspicious object detection, vehicle and vessel detection and tracking, vehicle attribute extraction, and number plate detection. These capabilities have been validated across multiple scenarios, including beach surveillance, sports venue monitoring, and national highway observation. The system also provides a clear pathway for future enhancement, including advanced behavioural analysis and multi-drone fusion, subject to operational validation and policy approval.

Security and access control have been treated as integral components of the design. Authentication mechanisms ensure that only authorised users can access the system. Role-based access control defines what functions a user may perform, while attribute-based access control evaluates contextual factors such as duty assignment, location, time, and case association before granting access. Tokens used for system access are short-lived, continuously validated, and can be revoked immediately in response to policy

changes or security incidents. All access decisions and revocations are logged in an immutable audit system, providing strong safeguards against misuse and legal risk.

The document also outlines a phased rollout roadmap to ensure responsible adoption. Initial pilot deployments focus on validating core functionality, operational procedures, and legal compliance. Subsequent phases expand coverage, introduce shared infrastructure, and formalise governance structures. Advanced capabilities are considered only after sufficient operational experience and policy review. This phased approach minimises risk and ensures that the system evolves in alignment with organisational readiness and legal oversight.

In summary, the proposed Drone Video Analytics System offers Kerala Police a robust, scalable, and legally sound platform for enhancing situational awareness and operational effectiveness. By combining advanced analytics with strong governance, evidence integrity, and human oversight, the system provides a practical pathway for the responsible use of drone technology in policing. It is designed not merely to showcase technology, but to deliver operational value while upholding the principles of accountability, proportionality, and rule of law.

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1. Introduction

The increasing use of unmanned aerial vehicles in policing has significantly enhanced situational awareness during public events, traffic management, disaster response, and law-and-order operations. Drones provide a wide-area aerial perspective that is difficult to achieve through fixed cameras or ground patrols alone. However, the large volume of video generated by multiple drones creates operational challenges in terms of real-time monitoring, timely interpretation, evidence handling, and long-term storage.

To address these challenges, Kerala Police requires a structured, scalable, and legally compliant drone video analytics system that converts raw aerial video into actionable operational intelligence without compromising privacy, security, or evidentiary integrity. The proposed system is designed as a decision-support platform that assists officers in understanding situations before, during, and after an event, while ensuring that final decisions remain with human authorities.

This document presents a comprehensive technical, operational, and legal framework for such a system, covering architecture, workflows, analytics capabilities, access control, data management, deployment strategy, and compliance considerations.

1.1 Background and Context

Kerala Police has progressively adopted technology-driven approaches to improve public safety, traffic regulation, crowd control, and emergency response. The deployment of drones for aerial surveillance and monitoring has already demonstrated value in managing large gatherings, monitoring road conditions, and supporting search and rescue operations.

Despite these advantages, current drone operations rely heavily on manual observation of live video feeds. This approach is not scalable when multiple drones are deployed simultaneously across different locations. Human operators may miss critical events due to cognitive overload, and continuous recording of raw video raises concerns related to storage costs, privacy, and legal defensibility.

At the same time, advances in computer vision and artificial intelligence have made it possible to automatically detect people, vehicles, crowds, and incidents in video streams. However, applying such technologies directly on drones or indiscriminately storing all video data is neither practical nor advisable in a policing context.

The proposed drone video analytics system has been conceived to bridge this gap. It combines lightweight edge processing, centralised GPU-based analytics, and controlled evidence management within an on-premise infrastructure owned and operated by the police. The system is designed to support multi-drone operations, provide timely alerts, generate verifiable digital evidence, and comply with Indian legal and constitutional requirements.

1.2 Objectives of the Drone Video Analytics System

The primary objective of the drone video analytics system is to enhance operational effectiveness of Kerala Police by transforming drone video feeds into meaningful, actionable intelligence while maintaining strict legal and ethical safeguards.

Specifically, the system aims to provide real-time and near-real-time insights related to crowd behaviour, traffic conditions, public safety incidents, and suspicious activities across multiple locations. It seeks to reduce reliance on continuous manual monitoring by assisting officers with automated alerts and structured visualisation.

Another key objective is to ensure that evidence generated from drone operations is reliable, traceable, and admissible. The system is designed to preserve chain of custody, prevent unauthorised access or tampering, and maintain detailed audit logs for all evidence-related actions.

The system also aims to be scalable and future-ready, capable of supporting an increasing number of drones and analytics capabilities without fundamental redesign. By adopting an on-premise deployment model and open-source components where feasible, the system ensures data sovereignty, cost efficiency, and long-term sustainability.

1.3 Scope of Deployment

The scope of this deployment covers the end-to-end lifecycle of drone video analytics, from video capture at the drone to intelligence visualisation and evidence retrieval at the central command level. It includes edge-level processing on operator laptops, central ingest and streaming infrastructure, GPU-based analytics at headquarters, secure evidence storage, structured databases, and multiple role-based dashboards.

The system is intended for use in a variety of policing scenarios, including crowd monitoring during public events, traffic and highway surveillance, disaster and emergency response, coastal and water-body monitoring, and incident investigation support. It is designed to operate with multiple drones concurrently, potentially deployed across different districts.

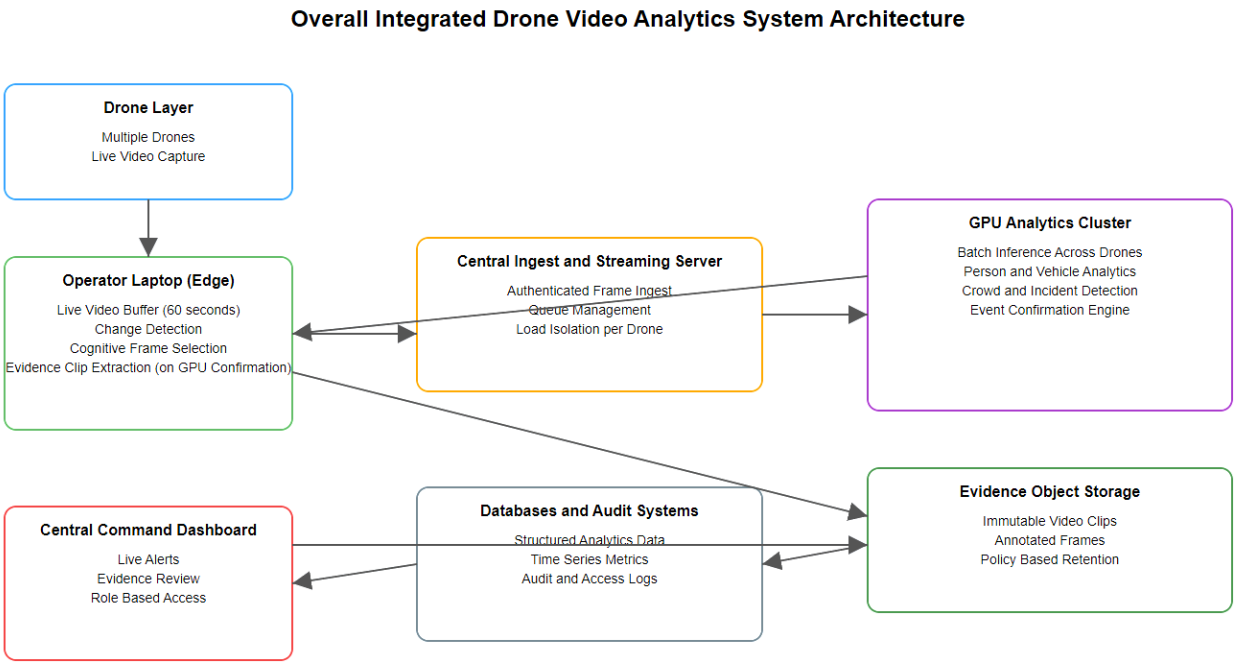
The deployment scope explicitly excludes autonomous enforcement actions. The system does not replace human judgement or decision-making. All analytics outputs are advisory in nature and are intended to support officers in making informed decisions in accordance with existing laws and procedures.

2. System Overview

The drone video analytics system is designed as a distributed, on-premise operational intelligence platform that processes aerial video streams from multiple drones and converts them into actionable insights and legally verifiable digital evidence. The system is not a monolithic application but a structured ecosystem in which sensing, pre-processing, analytics, evidence generation, storage, and access are deliberately separated. This clear separation of responsibilities ensures operational stability, scalability across multiple drones, and strong legal defensibility of all outputs.

At a high level, live video captured by drones is first processed at the edge on operator laptops, where short-term buffering and lightweight analysis are performed. Selected data is then transmitted through a central ingest and streaming layer to a GPU-based analytics cluster located at police headquarters. The analytics layer performs deep inference and confirms events of interest. Evidence creation and long-term storage are handled through secure, policy-controlled systems, while authorised users access results through role-based dashboards.

Each layer of the system has a clearly defined function and interacts with other layers only through controlled and authenticated interfaces. No single component performs end-to-end processing, and no layer has unnecessary access to raw video, analytics results, or stored evidence. This design minimises operational risk, prevents misuse, and enables the system to scale from pilot deployments to multi-district operations without architectural changes.



2.1 High-Level System Concept

The core concept of the system is to strike a balance between purely manual monitoring and indiscriminate automated surveillance. Rather than streaming all raw drone video to a central location or relying entirely on human observation, the system introduces intelligence at carefully chosen points in the processing pipeline. This ensures that technology assists officers without overwhelming infrastructure, personnel, or legal safeguards.

Live video captured by drones is first handled at the operator laptop, which serves as the edge layer of the system. At this stage, the video is buffered for a short duration and processed using lightweight techniques such as change detection and cognitive frame selection. These methods are designed to identify moments that are potentially relevant while preserving contextual continuity. Only representative and meaningful frames are forwarded for deeper analysis, significantly reducing bandwidth consumption and avoiding unnecessary transmission of routine or redundant data.

Centralised GPU resources located at police headquarters perform the primary analytics across all active drones. Frames received from the edge are processed in batches, allowing the system to efficiently utilise shared GPU capacity while maintaining near real-time performance. The analytics cluster performs computationally intensive tasks and generates confirmed events with associated confidence levels. These confirmed events are treated as authoritative and form the basis for alerts, operational awareness, and subsequent evidence handling.

Evidence is generated only after an event has been confirmed by central analytics and is extracted from the edge buffer to ensure the inclusion of pre-event and post-event context. The resulting evidence clips are stored in a dedicated on-premise storage system designed to ensure durability, integrity, and policy-based retention. Access to analytics outputs and evidence is provided through dashboards that are tailored to specific operational roles, ensuring that officers at different levels receive only the information relevant to their responsibilities.

2.2 Key Design Principles

The system is fundamentally designed around the principle of clear separation of responsibilities. Each layer of the architecture performs a specific and limited function: drones are used exclusively for video capture, operator laptops handle buffering and lightweight pre-processing, ingest servers manage secure data flow,

GPU clusters perform computationally intensive analytics, and storage systems are responsible for preserving evidence. This deliberate separation reduces system complexity, avoids functional overlap, and improves operational reliability and maintainability.

Data minimisation is another core principle guiding the design. The system does not retain continuous or indiscriminate surveillance footage. Video data is buffered temporarily at the edge and discarded unless an event is confirmed. Only evidence that is operationally relevant and legally justifiable is retained. This approach reduces privacy risks, limits storage overhead, and aligns with established legal and constitutional expectations regarding proportional surveillance.

Scalability has been built into the architecture from the outset. The system supports multiple drones operating concurrently and allows computing, storage, and networking resources to be expanded incrementally. Additional drones, ingest capacity, GPU resources, or storage can be added without redesigning the system. This enables phased deployment, controlled growth, and long-term sustainability as operational requirements evolve.

Security and access control are embedded as foundational elements rather than added as afterthoughts. Strong authentication mechanisms, role-based access control, attribute-based access control, and comprehensive audit logging are enforced consistently across all system components. These measures ensure accountability, enable traceability of all actions, and significantly reduce the risk of misuse or unauthorised access.

Finally, the system prioritises on-premise deployment and the use of open standards. This design choice ensures data sovereignty, reduces dependence on external service providers, and allows the Kerala Police to retain full administrative and operational control over infrastructure and data. It also facilitates integration with existing systems and long-term flexibility in technology choices.

2.3 Operational Philosophy (Decision Support, Not Automation)

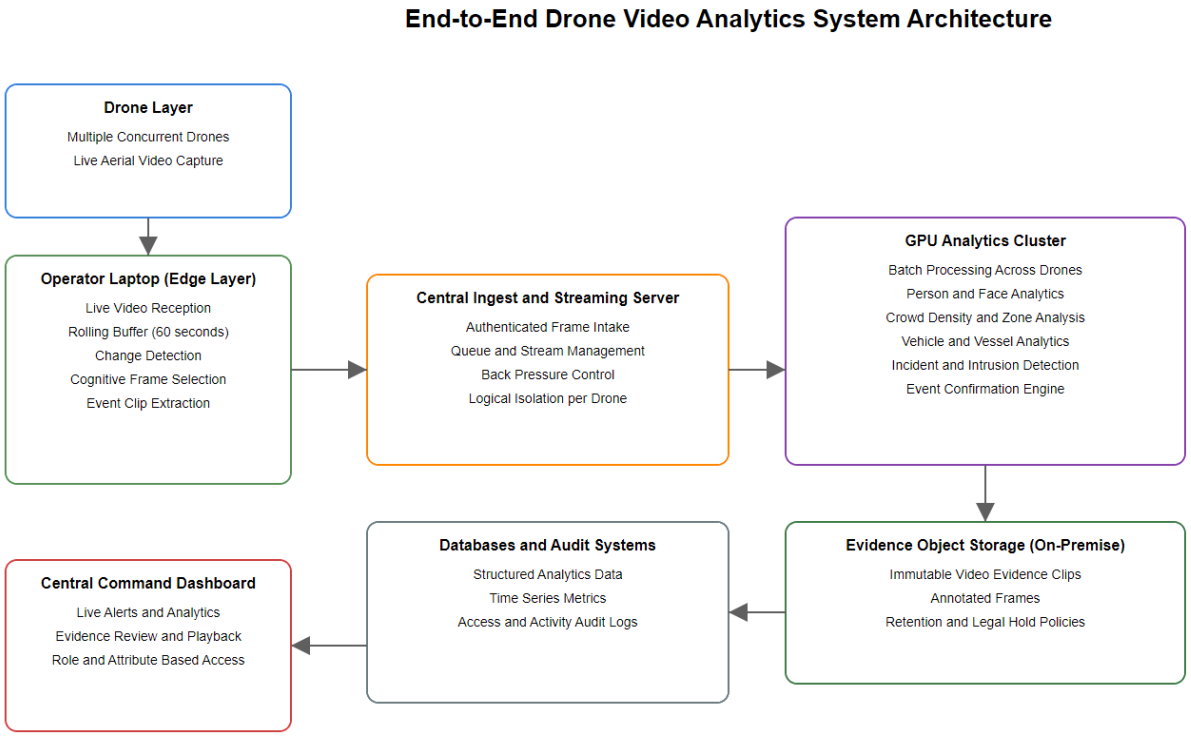
The operational philosophy of the system is firmly grounded in a human-in-the-loop approach to policing. The system is not designed to make enforcement decisions, initiate actions, or operate autonomously. Its role is to support officers by processing large volumes of visual data and presenting structured information that may assist in situational assessment and decision-making.

Analytics outputs generated by the system, such as crowd density estimates, vehicle movement patterns, intrusion alerts, or incident indicators, are presented as informational signals rather than commands. These outputs highlight conditions or events that may warrant attention, but they do not prescribe responses. Officers remain fully responsible for interpreting system outputs in conjunction with on-ground observations, intelligence inputs, operational directives, and supervisory judgement.

By adhering to this philosophy, the system enhances situational awareness while preserving human authority and accountability. It prevents over-reliance on automated interpretations and ensures that operational decisions are always made by trained personnel. This approach supports ethical use of technology, mitigates the risk of inappropriate automation, and aligns with constitutional principles and legal expectations governing the deployment of surveillance technologies in policing.

3. End-to-End System Architecture

The end-to-end system architecture defines how drone video data flows from capture to analytics, evidence storage, and operational decision-making. The architecture follows a layered and modular design in which each component has a clearly defined responsibility. This design ensures scalability, security, legal defensibility, and operational clarity while supporting simultaneous operation of multiple drones across different locations.



The overall architecture integrates drone hardware, edge processing, central ingest infrastructure, GPU-based analytics, secure evidence storage, structured databases, access control mechanisms, and role-based dashboards into a single coherent system.

3.1 Overall Integrated Architecture

At a high level, the system is organised around a central analytics and command core, supported by distributed edge components and tightly controlled data pipelines. The architecture deliberately separates sensing, processing, decision support, storage, and access, ensuring that each function is performed at the most appropriate layer from both operational and legal perspectives.

Drones operate strictly as sensing devices. They capture live video streams but do not perform analytics, decision-making, or long-term storage. This design choice keeps drones lightweight, reduces risk at the point of capture, and ensures that no sensitive processing or data retention occurs at the aerial layer.

Operator laptops form the edge processing layer. Each laptop maintains a temporary rolling buffer of live video and performs lightweight operations such as change detection and cognitive frame selection. These functions are designed to reduce bandwidth usage and latency without making any enforcement or evidentiary decisions. The edge layer also listens asynchronously for confirmed event signals from the central analytics layer, enabling controlled evidence clip extraction when required.

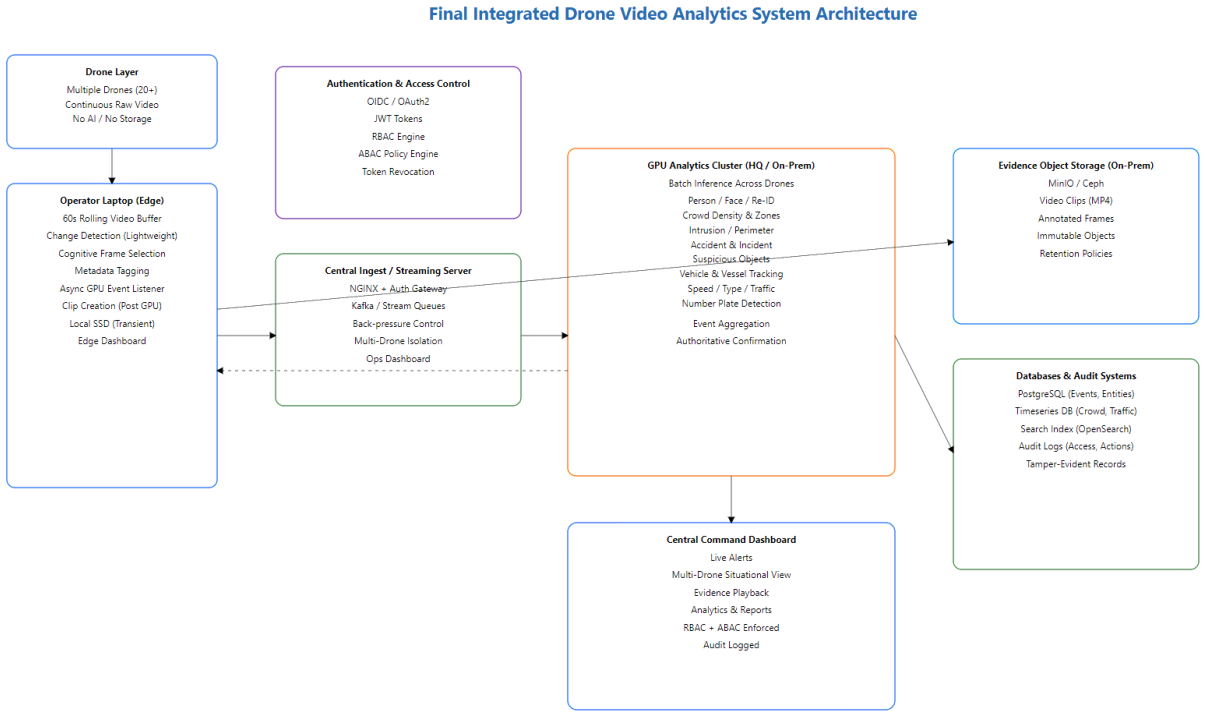
A central ingest and streaming server governs all data flow between the edge and the analytics core. This layer authenticates incoming streams, enforces access control, manages queues, applies back-pressure, and isolates multiple drone feeds from one another. By acting as a controlled gateway, the ingest server ensures system stability even when many drones are active simultaneously.

The GPU analytics cluster at headquarters performs all computationally intensive analytics and acts as the authoritative source for event confirmation. Analytics such as person and vehicle tracking, crowd density estimation, intrusion detection, and incident analysis are executed here using batched processing across multiple drone feeds. Only events confirmed at this layer are treated as operationally valid.

Evidence generated from confirmed events is stored in a dedicated on-premise object storage system designed for immutability, retention enforcement, and legal defensibility. Structured analytics results and system metadata are stored separately in databases optimised for querying, time-series analysis, and audit logging. This separation ensures both performance and compliance.

Access to all components is governed by a central authentication and access control framework implementing role-based and attribute-based controls, with comprehensive audit logging. Operational visibility is provided through role-specific dashboards that present alerts, analytics, and evidence without exposing raw or unauthorised data.

This integrated architecture avoids single points of failure, prevents uncontrolled data propagation, and ensures that operational efficiency, legal defensibility, and accountability are addressed coherently at every layer of the system.



3.2 Drone Layer

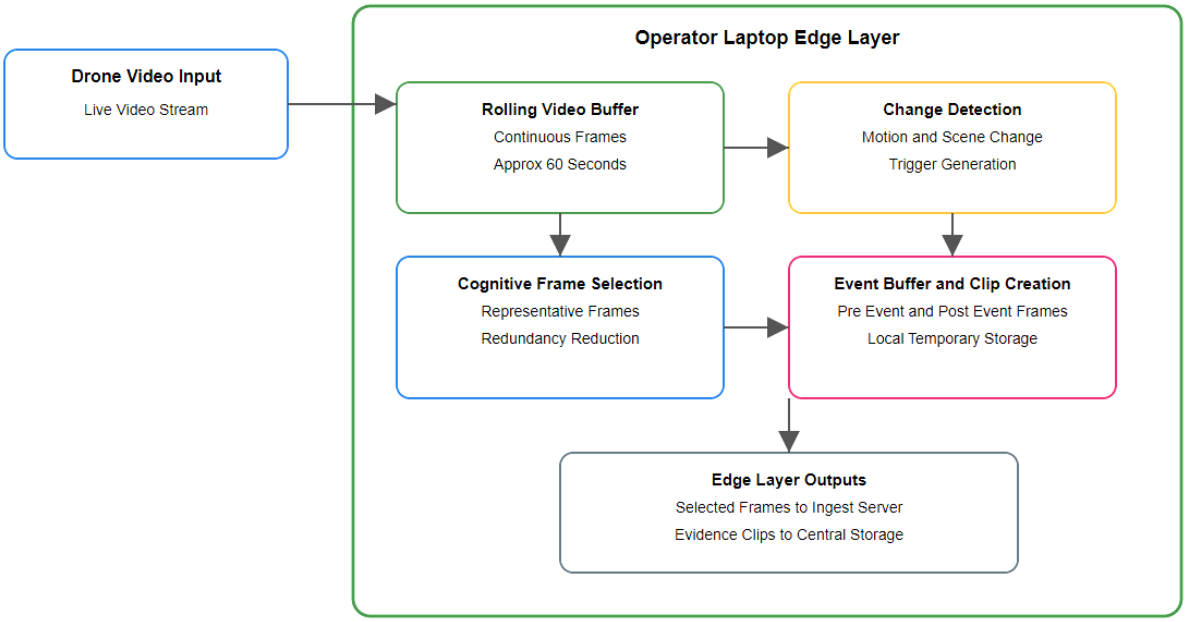
The drone layer consists of multiple drones, typically twenty or more, operating concurrently across different geographic locations. Each drone captures continuous raw video using onboard cameras and transmits this video to the assigned operator laptop.

Drones do not perform any artificial intelligence processing, analytics, or storage. This ensures that drone hardware remains lightweight, power-efficient, and compliant with aviation regulations. By restricting drones to capture and transmission only, the system avoids complexity and ensures that all intelligence decisions are taken within controlled, auditable environments.

3.3 Operator Laptop (Edge Layer)

The operator laptop is the first intelligent processing layer in the system. It receives the live video stream from the drone and maintains a rolling video buffer of approximately sixty seconds. This buffer exists solely to support event context and is continuously overwritten.

Operator Laptop Edge Processing Architecture



Within this layer, lightweight change detection algorithms monitor the video stream to identify moments of potential interest. Cognitive frame selection then chooses representative frames that best describe the current scene. These selected frames, along with associated metadata, are forwarded to the central ingest server for deeper analysis.

The operator laptop does not perform final analytics or make enforcement decisions. It listens asynchronously for event confirmations from the GPU analytics cluster. When a confirmed event is received, the laptop extracts the relevant video segment from its rolling buffer and creates an evidence clip. This clip is temporarily stored on local solid-state storage and then uploaded to central evidence storage.

An edge dashboard on the laptop provides the drone operator with visibility into stream status, buffer health, and system connectivity, without exposing analytics results or sensitive evidence.

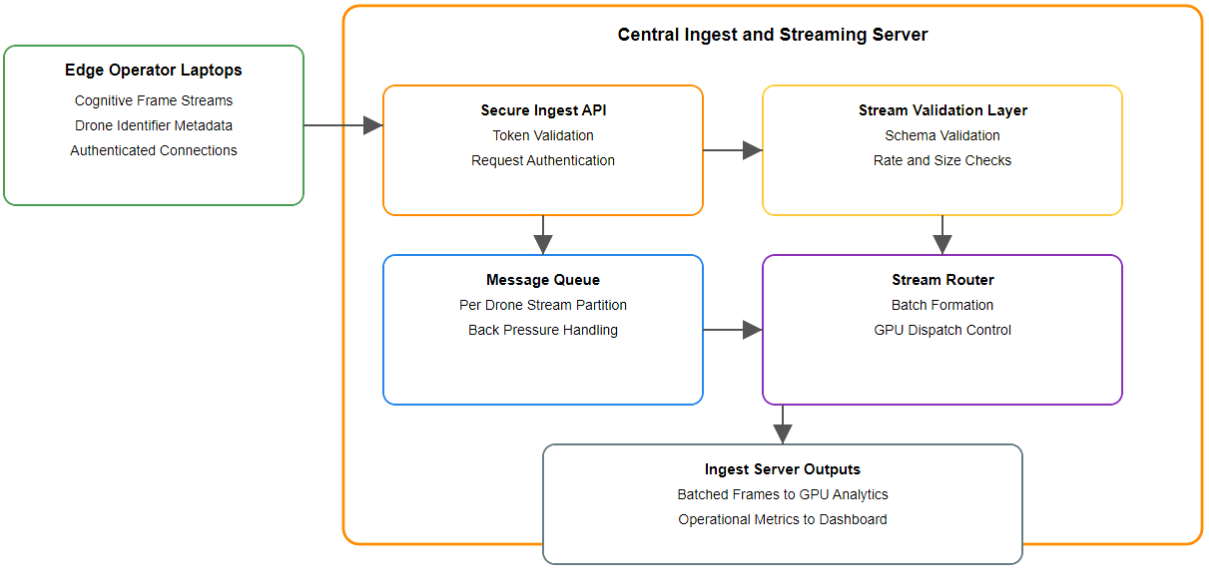
3.4 Central Ingest and Streaming Server

The central ingest and streaming server acts as the controlled gateway between edge devices and the analytics infrastructure. It receives cognitively selected frames from multiple operator laptops through authenticated channels.

This server performs stream validation, enforces rate limits, manages back-pressure, and queues incoming frames using a streaming or messaging backbone. Logical isolation between drone feeds is maintained to prevent interference and ensure predictable performance. The ingest server does not store evidence or raw video and does not perform analytics.

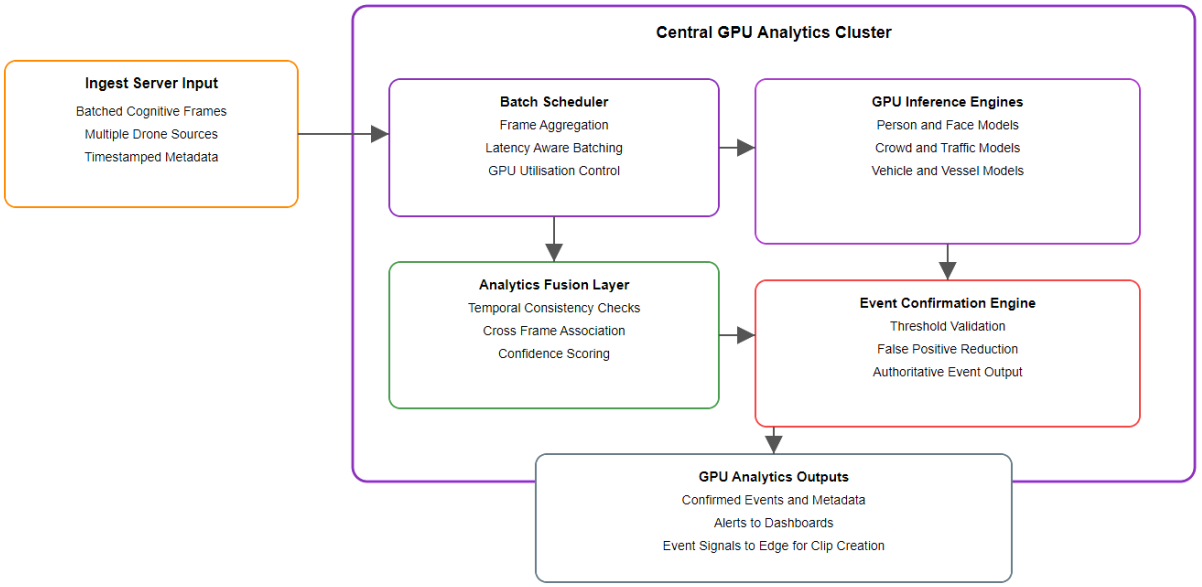
An operations dashboard provides system administrators with visibility into feed health, queue depth, throughput, and system alerts. This dashboard is strictly operational and does not allow access to video content or analytics results.

Central Ingest and Streaming Server Architecture



3.5 GPU Analytics Cluster

GPU Analytics Cluster for Multi Drone Batch Processing

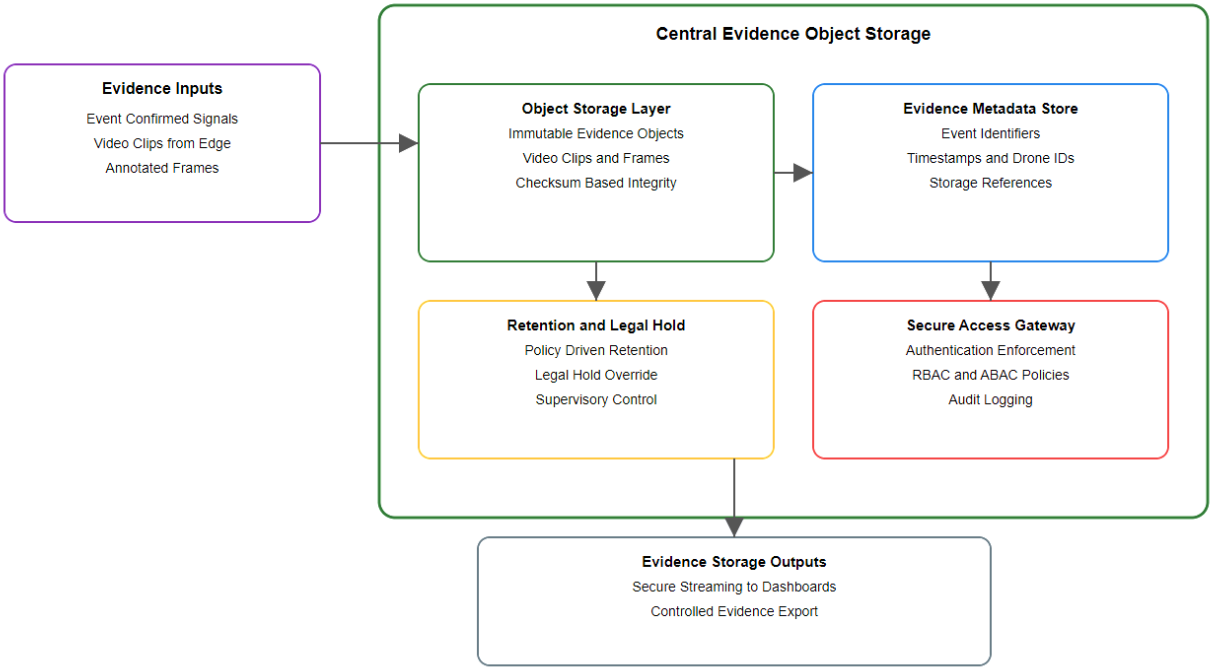


The GPU analytics cluster, hosted on-premise at headquarters, is the core intelligence layer of the system. It receives queued frames from the ingest server and processes them in batches across multiple drones to maximise GPU utilisation. The cluster executes all analytics capabilities, including person detection and re-identification, face analysis, crowd density estimation and zoning, intrusion and perimeter breach detection, accident and incident detection, suspicious object detection, vehicle and vessel tracking, speed and traffic analysis, and number plate detection. Analytics outputs are aggregated and evaluated to produce confirmed events with confidence scores. Only these confirmed events are treated as authoritative. The cluster does not store video clips and does not interact directly with dashboards.

3.6 Evidence Storage System

Confirmed evidence video clips and annotated frames are stored in a dedicated on-premise object storage system. This storage layer is implemented using open-source object storage platforms such as MinIO or Ceph and is designed for durability, integrity, and controlled access.

On Premise Evidence Storage System Architecture

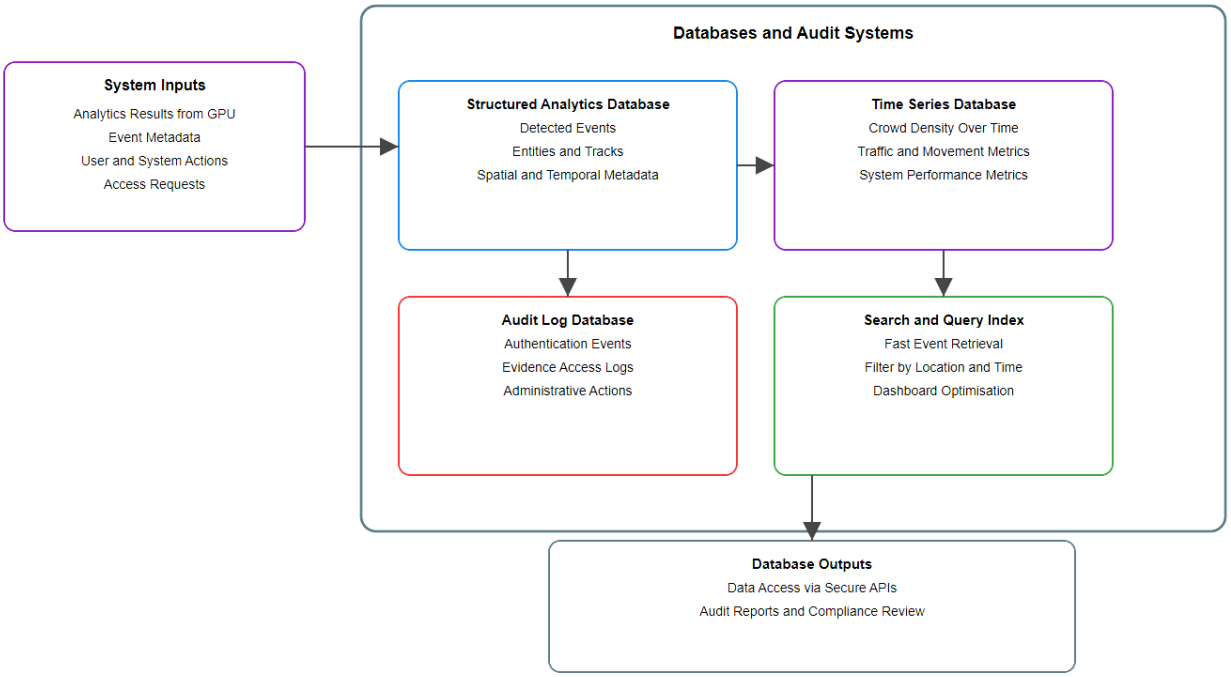


Evidence objects are stored in immutable form and are governed by retention policies defined by operational and legal requirements. The storage system is isolated from ingest and analytics infrastructure to preserve chain of custody and prevent accidental or unauthorised modification. Access to evidence is mediated exclusively through backend APIs that enforce authentication, role-based access control, and attribute-based access control.

3.7 Databases and Audit Systems

Structured analytics outputs and system metadata are stored in dedicated databases. Relational databases store events, detected entities, trajectories, and associations. Time-series databases capture temporal analytics such as crowd density and traffic patterns. Search indexes support efficient retrieval and filtering of records. Audit systems record all access, actions, and system decisions, including evidence creation, viewing, export, and administrative operations. Audit records are tamper-evident and retained for extended periods to support internal reviews, legal scrutiny, and compliance audits.

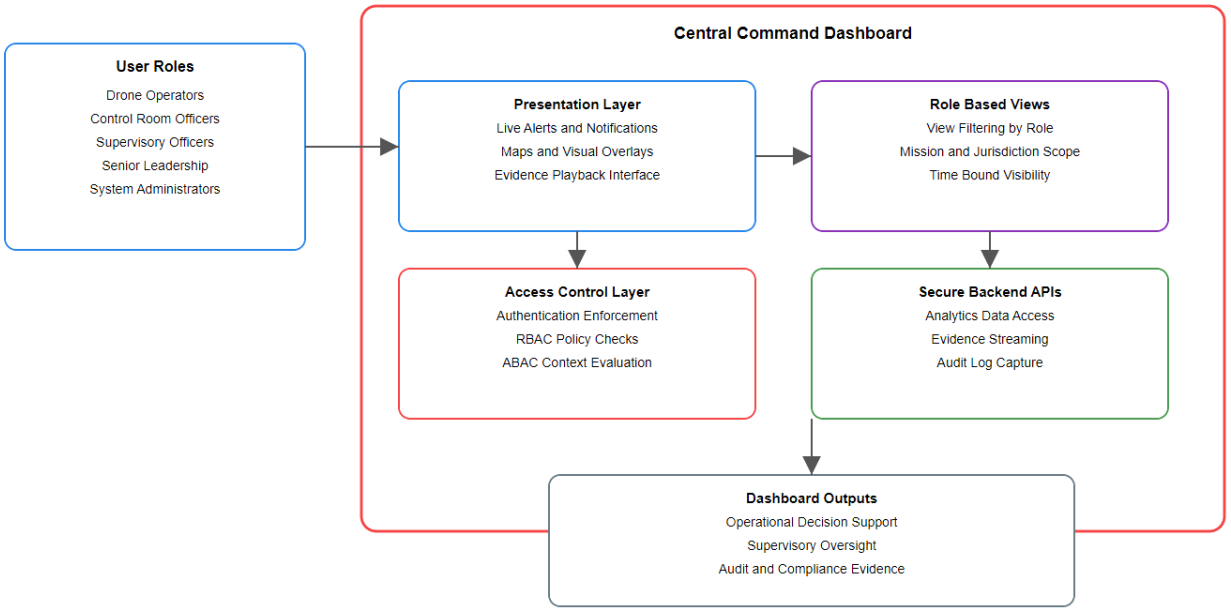
Structured Databases and Audit Logging Architecture



3.8 Central Command Dashboard

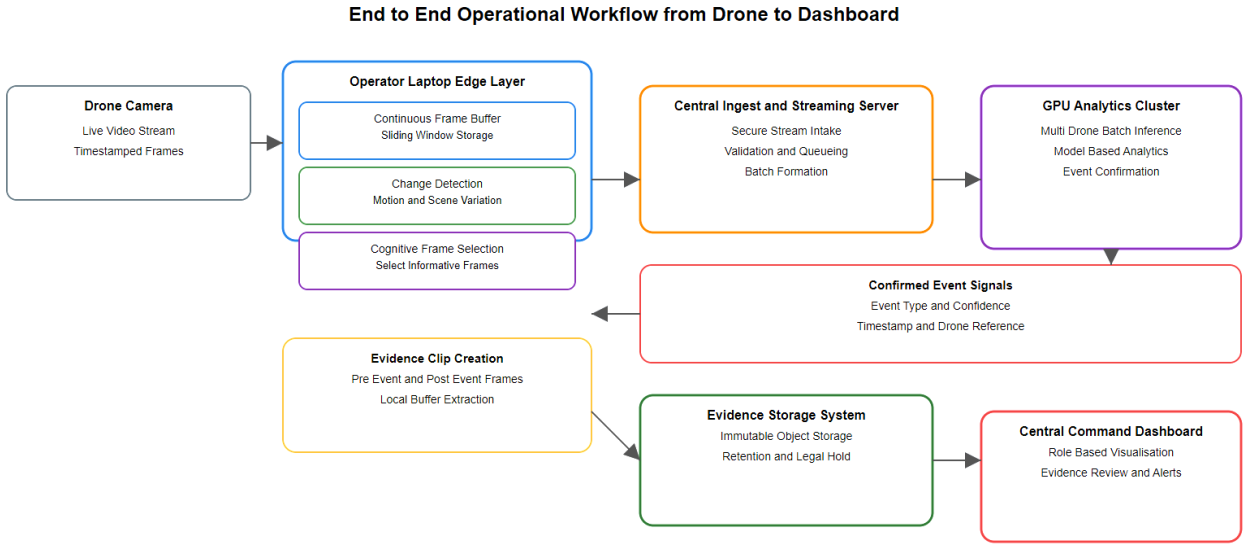
The central command dashboard is the primary operational interface for control room officers, supervisors, and senior leadership. It presents live alerts, multi-drone situational views, analytics summaries, and evidence playback. All dashboard access is governed by authentication, RBAC, and ABAC policies. Users see only data relevant to their role, mission, location, and time window. Every interaction with the dashboard is logged for audit purposes. The dashboard supports informed decision-making while ensuring that authority and accountability remain with human officers.

Central Command Dashboard Layers and Access Separation



4. Operational Workflow

The operational workflow describes how video data moves through the system from the moment it is captured by a drone until it is reviewed by authorised officers through dashboards. The workflow is designed to support real-time operations while ensuring that data handling remains controlled, auditable, and legally defensible.



The workflow deliberately separates transient processing from permanent storage and separates advisory analytics from human decision-making. Each step is executed by a specific system component with a clearly defined responsibility.

4.1 Video Capture and Edge Buffering

Each drone continuously captures raw video during operation and transmits the live feed to its assigned operator laptop. The drone itself does not store video or performs analytics. Upon receiving the video stream, the operator laptop maintains a rolling buffer of approximately sixty seconds. This buffer continuously overwrites older frames and exists solely to provide temporal context around potential incidents. At no point is the full continuous video stream permanently stored. The buffering mechanism is timestamp-indexed, allowing the system to accurately extract video segments corresponding to specific time windows when required. This design ensures that short delays in downstream processing do not result in loss of contextual information.

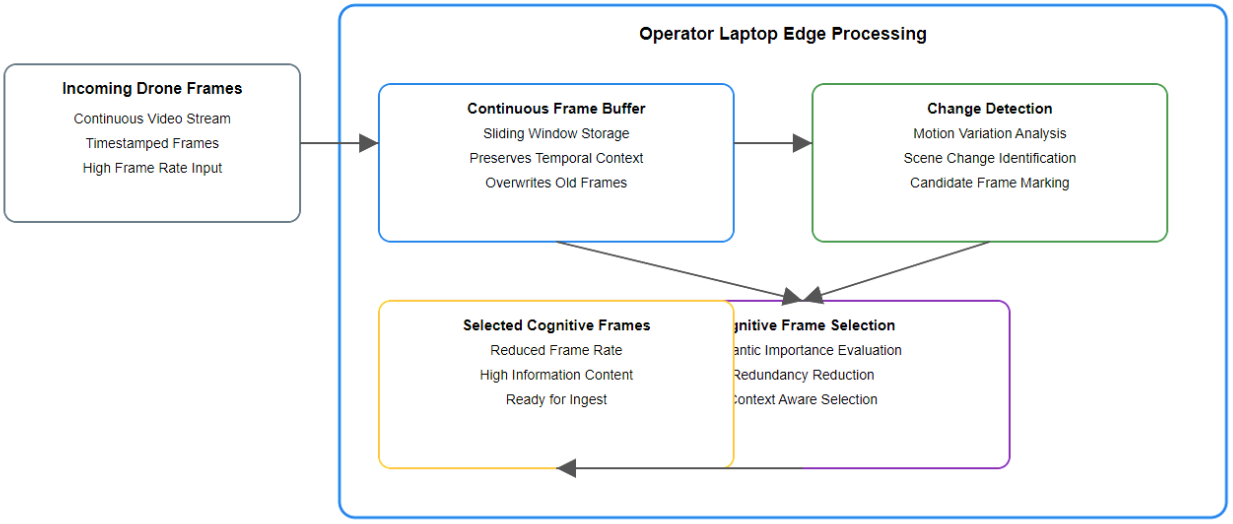
4.2 Change Detection and Cognitive Frame Selection

Within the operator laptop, lightweight change detection operates continuously on the buffered video stream. The purpose of change detection is to identify moments where the scene content changes meaningfully, such as sudden crowd movement, appearance of vehicles, or entry into restricted zones. Change detection functions as a trigger mechanism and does not make decisions about incidents.

Cognitive frame selection operates as a separate process. Its purpose is to select a small number of representative frames that best describe the current scene, considering factors such as motion, scene diversity, and redundancy reduction. Cognitive frame selection determines *what information is sent for deeper analysis*, not *whether an incident has occurred*.

The selected frames, along with associated metadata such as drone ID, timestamp, and location, are forwarded to the central ingest server. This approach reduces bandwidth usage, improves scalability, and limits unnecessary data propagation.

Change Detection and Cognitive Frame Selection at Edge Layer



Concrete Operational Example: Beach Surveillance Scenario

Consider a drone monitoring a beach area during peak hours. The drone camera continuously captures video at around 30 frames per second. Most of these frames are visually very similar — people walking slowly, waves moving, shadows changing slightly, and small camera vibrations due to wind. At the operator laptop, **all incoming frames are first stored in a rolling buffer**, which always keeps, for example, the last 60 seconds of video. This buffer exists only to preserve time continuity; it does not decide what is important.

Now, **change detection** runs continuously on this buffered stream. Its job is very simple and very fast. It does not understand people, behaviour, or intent. It only asks: *"Is something changing compared to recent frames?"*

For example:

- A sudden increase in motion in one zone
- A group that was stationary starts moving together
- A previously empty area suddenly has activity
- A vehicle enters a pedestrian-only zone

When such changes are observed, change detection **marks candidate timestamps**. It does not decide whether the change is meaningful or suspicious. It only says: *"Something changed here."*

Next, **cognitive frame selection** takes over. It looks at the frames around those marked timestamps **and also at neighbouring frames from the buffer**. This stage uses higher-level logic and lightweight AI cues. It asks questions like:

- Are people involved, or is it just waves or shadows?
- Is the change persistent or momentary?
- Is this scene different from previous patterns in this area?
- Does this frame add new information compared to the last selected frame?

From possibly hundreds of frames flagged by change detection, cognitive frame selection may choose only a small number — for example, 15 to 20 frames per minute — that best represent what is happening.

These selected frames are then sent forward to the central ingest server and GPU for detailed analytics. **The rest of the frames remain only in the local buffer and are never transmitted.**

Why Both Are Needed

If we used **only change detection**, the system would send too many frames whenever there is wind, camera motion, or crowd movement, overwhelming the network and GPU. If we used **only cognitive frame selection without change detection**, the system would have to analyse every frame semantically, which is computationally expensive and unnecessary at the edge.

By separating them:

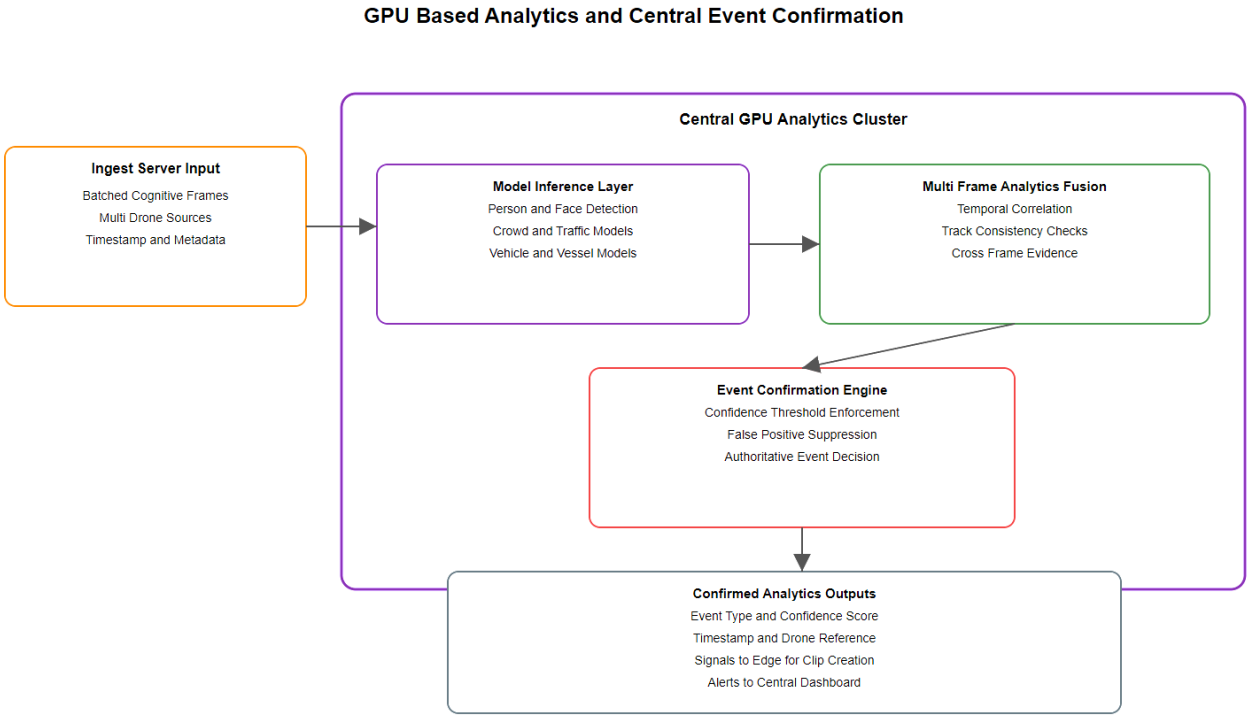
- Change detection acts as a **cheap early warning filter**
- Cognitive frame selection acts as an **intelligent summariser**
- The rolling buffer ensures **temporal context is never lost**

This separation is what allows the system to scale from one drone to twenty drones without flooding the network or GPU.

4.3 Ingest and Streaming Workflow

The central ingest and streaming server receives cognitively selected frames from multiple operator laptops through authenticated and encrypted channels. Each incoming stream is validated and associated with the correct drone and mission context. The ingest server queues incoming frames using a streaming or messaging backbone and applies back-pressure controls to ensure system stability during peak load. Logical isolation between drone feeds is maintained to prevent one feed from affecting others. The ingest server does not store raw video or evidence. Its sole function is to regulate data flow into the analytics layer and provide operational visibility to system administrators through an operations dashboard.

4.4 GPU-Based Analytics and Event Confirmation



The GPU analytics cluster retrieves frames from the ingest queues and processes them in batches across multiple drones. Batch processing enables efficient use of GPU resources while maintaining near real-time responsiveness.

The analytics cluster applies all configured analytics models, including detection, tracking, estimation, and classification functions. Analytics outputs are aggregated across time and frames to determine whether a confirmed event has occurred.

Only events that meet defined confidence and consistency criteria are treated as confirmed. These confirmed events are considered authoritative and are the sole triggers for downstream evidence generation and alerts. The GPU cluster does not create video clips and does not store video data.

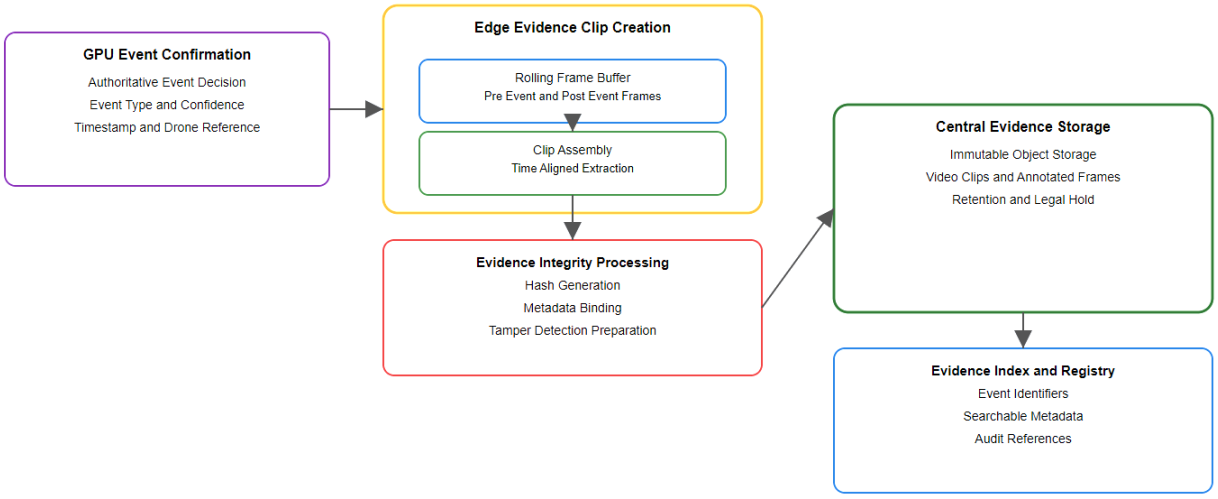
At the GPU layer, analytics are no longer frame-by-frame guesses. Multiple frames from multiple drones are analysed together, correlated over time, and validated against confidence thresholds. Only after this fusion and validation does the system declare an event. This makes the GPU the authoritative decision point, ensuring consistency, reducing false alarms, and maintaining evidentiary reliability."

4.5 Evidence Clip Creation and Upload

When a confirmed event is generated by the GPU analytics cluster, an event notification containing relevant timestamps and identifiers is sent asynchronously to the corresponding operator laptop.

Upon receiving this notification, the operator laptop extracts a continuous video segment from its rolling buffer, including a defined pre-event and post-event window. This ensures that the evidence clip contains sufficient context around the incident.

Evidence Clip Creation and Storage Pipeline



The extracted clip is encoded in a standard format and temporarily stored on local solid-state storage. It is then uploaded to the central on-premise evidence object storage through secure backend APIs. After successful upload, the local copy is discarded according to policy. This workflow ensures that evidence clips are complete, contextual, and generated only after authoritative confirmation.

Once the central GPU confirms an event, the local laptop extracts the relevant video from its buffered recording, creates a complete clip with pre-event and post-event context, applies integrity safeguards, and uploads it to secure on-premise storage where it becomes immutable evidence.

4.6 Dashboard Visualisation and Review

Confirmed events, analytics summaries, and evidence references become available through the central command dashboard. Authorised officers can view alerts, examine analytics results, and play evidence clips based on their assigned roles and permissions.

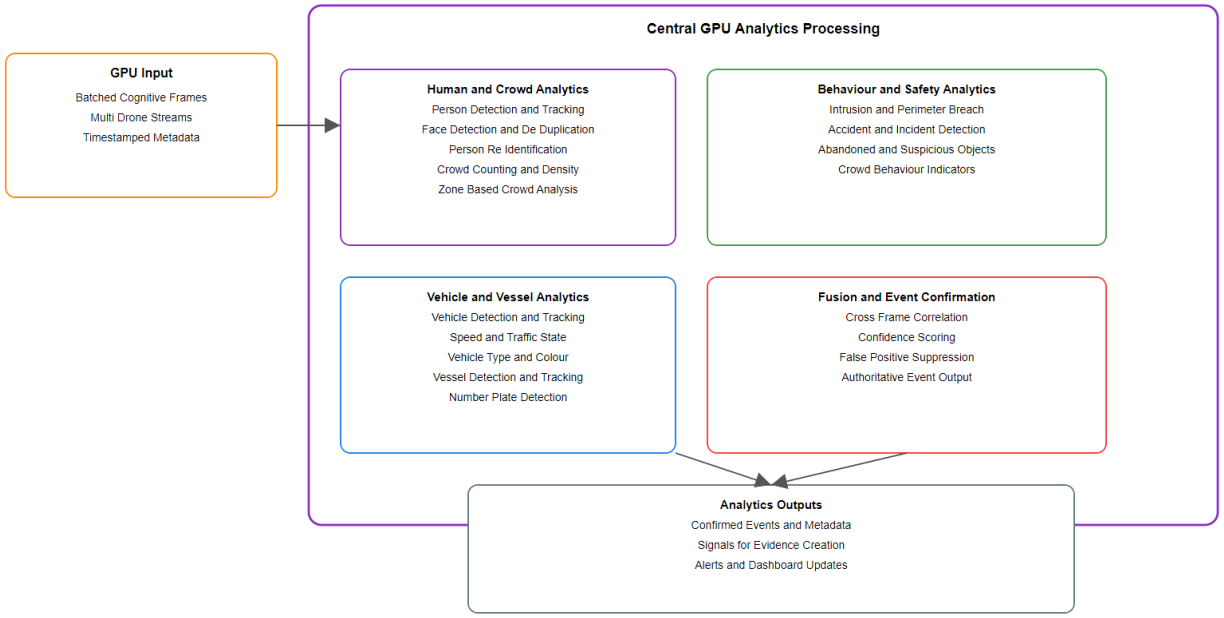
All dashboard access is governed by authentication, RBAC, and ABAC rules. Users see only the information relevant to their role, mission, location, and time window. Evidence playback is streamed securely from the object storage system using time-limited access mechanisms.

Every access, view, or action performed through the dashboard is logged in the audit system. This ensures accountability, traceability, and compliance with legal and administrative requirements.

5. Analytics Capabilities

The drone video analytics system provides a comprehensive set of computer vision and video analytics capabilities designed to support real-world policing scenarios. All analytics are executed centrally on the GPU analytics cluster and operate only on cognitively selected frames received from edge devices. The system is explicitly designed as a decision-support platform, where analytics outputs assist officers in understanding situations rather than initiating automated actions.

Mapping of Analytics Capabilities to GPU Processing



The analytics capabilities have been validated using multiple types of drone footage, including beach surveillance, sports hub monitoring, and national highway scenarios. Where applicable, limitations and data-dependency considerations are clearly stated.

5.1 Person Detection, Tracking and Re-Identification

The system supports robust detection of individuals in drone video footage, even under varying altitude, camera angle, and crowd density conditions. Once detected, individuals are tracked across consecutive frames using persistent identifiers. Re-identification capabilities allow the system to associate the same individual across non-contiguous frames and, where applicable, across multiple video segments or drone feeds. This enables continuity of observation without storing continuous video, supporting lawful monitoring during crowd movement, dispersal, or pursuit scenarios. Tracking and re-identification results are treated as analytical indicators and are always subject to human interpretation.

5.2 Face Detection and Image De-duplication

Face detection is performed on detected persons where facial features are sufficiently visible. The system extracts face regions and applies image de-duplication techniques to avoid redundant storage of identical or near-identical face images. This capability helps reduce data volume and improves efficiency during investigations. Face detection outputs are not treated as identity confirmation and are not used for autonomous identification. Any downstream use of face data remains subject to legal authorisation and procedural safeguards.

5.3 Crowd Detection and Density Estimation

The system supports estimation of crowd size and density using deep learning–based density mapping techniques. It can generate approximate counts and density heatmaps over time, even in scenarios where individual detection is challenging due to scale or occlusion. Crowd density estimates are particularly useful during public events, festivals, and emergency situations to identify congestion, overcrowding, or abnormal movement patterns. These outputs are advisory and intended to support situational awareness.

5.4 Zone Segmentation and Area-wise Analysis

The analytics engine supports segmentation of the observed area into predefined zones such as entry points, restricted areas, open grounds, or road segments. Crowd density, movement, and activity levels are computed separately for each zone. This zone-based analysis enables officers to understand not only overall conditions but also localised variations within the monitored area. It is especially useful for managing large venues, beaches, and urban intersections.

5.5 Intrusion and Perimeter Breach Detection

The system supports detection of intrusions into defined restricted or sensitive zones. Virtual perimeters can be configured based on operational requirements, and the analytics engine monitors for unauthorised entry or movement across these boundaries. Intrusion alerts are generated only after consistency checks to reduce false positives. These alerts assist officers in responding promptly to potential security breaches without relying on continuous manual monitoring.

5.6 Accident and Incident Detection

The analytics engine includes capabilities to identify potential accidents and incidents based on sudden changes in motion patterns, vehicle behaviour, or crowd reactions. This is particularly relevant for highway monitoring and large public gatherings. Detected incidents are flagged for human review and are not treated as confirmed without further validation. This ensures that analytics assist in early detection while maintaining operational caution.

5.7 Abandoned and Suspicious Object Detection

The system supports detection of objects that appear unattended or remain stationary beyond a configurable time threshold. This capability is data-dependent and performs best when background conditions are stable and camera motion is moderate. Suspicious object detection is intended as an early warning mechanism. Final assessment and response decisions remain with trained officers, who consider ground intelligence and contextual factors.

5.8 Vehicle Detection and Tracking

The analytics engine detects vehicles such as cars, buses, trucks, two-wheelers, and auto-rickshaws in drone footage. Detected vehicles are tracked over time using persistent identifiers, enabling trajectory analysis and movement history. Vehicle tracking supports monitoring of traffic flow, pursuit scenarios, and congestion patterns without storing continuous raw video.

5.9 Vehicle Attributes (Speed, Type, Colour, Traffic State)

For tracked vehicles, the system estimates attributes including approximate speed, vehicle type, and dominant colour. It also classifies vehicle states such as parked versus moving and assesses overall traffic conditions as normal or congested. These attributes provide situational context for traffic management and incident response. All estimates are indicative and subject to environmental and calibration factors.

5.10 Vessel Detection and Tracking

The system supports detection and tracking of vessels in coastal and inland water bodies, including boats and ferries. Vessel trajectories and movement patterns can be analysed to support coastal monitoring, event management, and safety operations. Vessel tracking capabilities have been validated using drone footage from beach and waterfront scenarios.

5.11 Number Plate Detection

Number plate detection is supported where plates are clearly visible in the video. This capability has been demonstrated on suitable datasets and controlled scenarios. Due to factors such as drone altitude, camera angle, and motion blur, number plate detection performance is context-dependent. The system does not attempt plate recognition where visibility conditions are insufficient.

5.12 Evidence Outputs and Forensic Logs

All confirmed analytics outputs result in structured evidence artefacts rather than raw video dumps. These include annotated video clips highlighting detected events, annotated frames for quick review, and structured forensic logs in CSV and JSON formats.

Forensic logs capture details such as timestamps, drone identifiers, event types, confidence scores, and system actions. These logs are stored in secure databases and are accessible through dashboards based on role and authorisation.

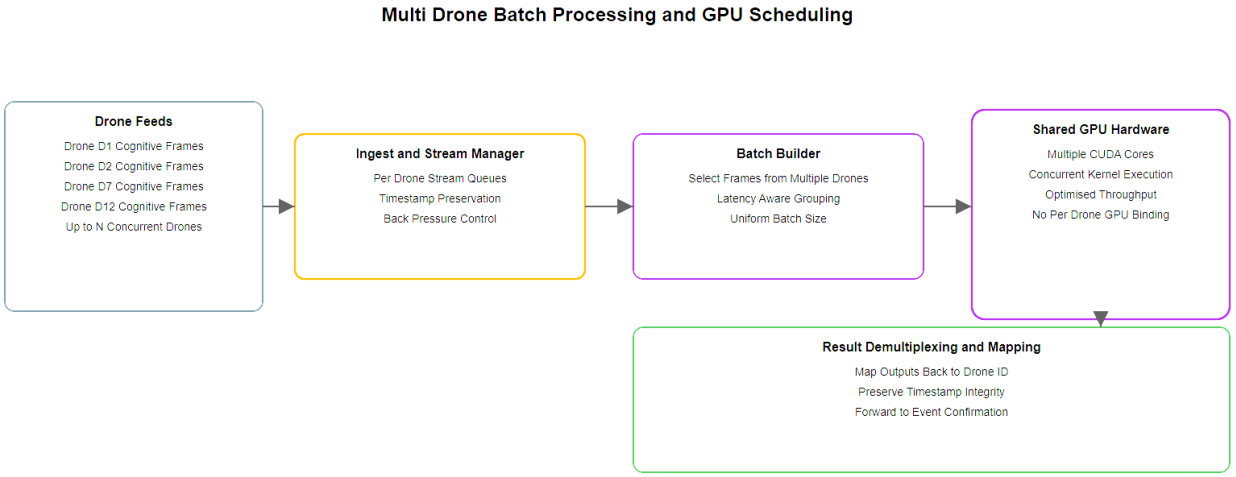
Dashboard views present analytics results, evidence playback, and summaries in a form suitable for operational review, reporting, and audit. All access and actions related to evidence are logged to maintain accountability and chain of custody.

Note on Advanced Capabilities: Advanced analytics such as crowd behaviour prediction in drone video, free-standing conversation group detection in drone footage, multi-drone fusion, cross-camera tracking, and altitude-adaptive behaviour models are identified as enhancement areas. These capabilities require further calibration using Kerala Police–specific data and operational feedback and are therefore not treated as baseline features in the current deployment.

6. Multi-Drone Processing and Scalability

The drone video analytics system is designed from the outset to support simultaneous operation of multiple drones across different locations. Multi-drone handling is treated as a core requirement rather than an extension, and all components are architected to scale predictably without compromising performance, stability, or legal safeguards. The system avoids per-drone isolation at the compute level and instead adopts

controlled aggregation and batching strategies that enable efficient resource utilisation while maintaining logical separation and traceability.



6.1 Handling Multiple Concurrent Drone Feeds

Each drone operates independently at the sensing level and is assigned a dedicated operator laptop. This ensures that video capture, buffering, and lightweight pre-processing are isolated at the edge and do not interfere with other drone operations. At the backend, all drone feeds converge at the central ingest and streaming server. Each feed is uniquely identified using drone identifiers, mission identifiers, and timestamps. Logical isolation is maintained through separate stream partitions or queues, preventing congestion or failure in one feed from affecting others. This design allows the system to handle multiple concurrent drone feeds in parallel while maintaining deterministic behaviour, predictable latency, and clear traceability of data from source to analytics output.

6.2 Batch Processing and GPU Scheduling

The GPU analytics cluster processes data from all drones using batch inference rather than allocating dedicated GPU resources per drone. Frames received from different drones are grouped into batches based on arrival time and processing priority.

For example, a single inference batch may contain frames from multiple drones captured at slightly different times. Each frame carries metadata identifying its drone, timestamp, and context. After inference, results are mapped back accurately to their originating drone and frame.

GPU scheduling is handled by the analytics orchestration layer and underlying deep learning frameworks. The system does not manually assign GPU cores to individual drones. Instead, it relies on batching, queue management, and workload balancing to achieve optimal GPU utilisation. This approach ensures scalability and avoids underutilisation or unnecessary hardware duplication.

6.3 Service-Oriented Analytics Architecture

Analytics capabilities are implemented as logically distinct services within the GPU analytics cluster. Each capability, such as person detection, crowd estimation, or vehicle tracking, operates as a modular processing stage within the overall analytics pipeline.

While these services are logically separate, they are not deployed as isolated GPU instances. Instead, they share GPU resources and are orchestrated as part of a unified inference workflow. This allows new analytics capabilities to be added or existing ones to be updated without disrupting the entire system.

The service-oriented design also supports selective enablement of analytics based on operational needs. For example, traffic analytics may be prioritised during highway monitoring, while crowd analytics may be emphasised during public events.

6.4 Scalability Considerations

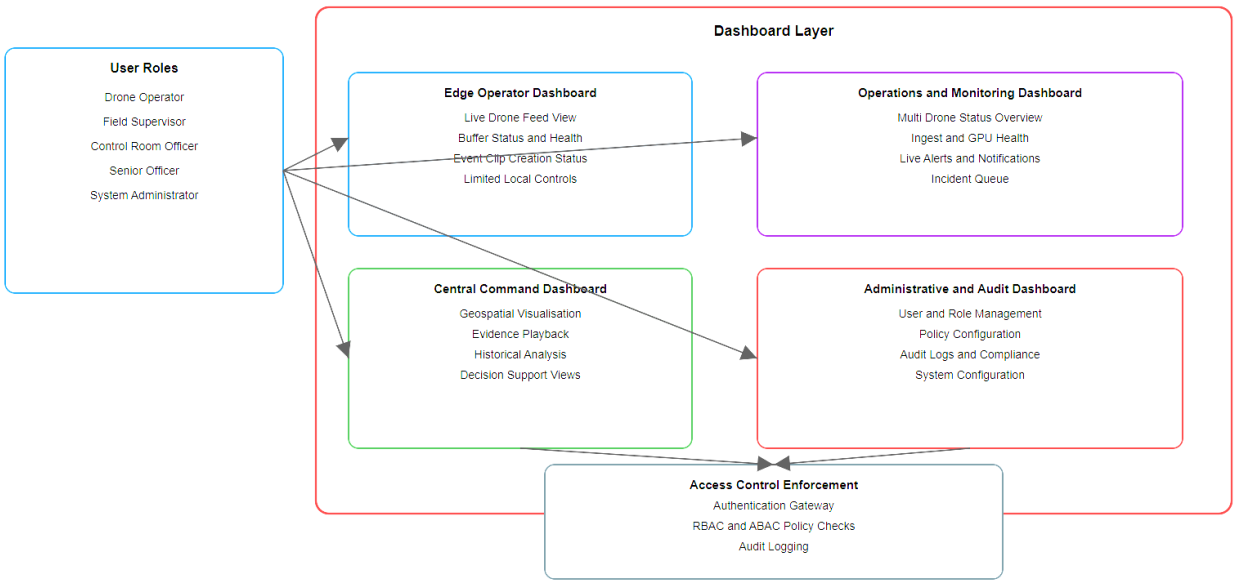
The system is designed to scale horizontally and vertically across all major components. Scaling at the edge involves adding more operator laptops as the number of drones increases. Scaling at the ingest layer involves increasing server capacity or adding additional ingest nodes if required.

GPU analytics scaling is achieved by adding additional GPUs or nodes to the analytics cluster. Because analytics are batch-based and centrally orchestrated, scaling does not require changes to drone or edge configurations. Evidence storage and databases scale independently by adding storage capacity or database replicas. Retention policies and archival strategies ensure that long-term growth remains manageable. This modular and layered scalability approach ensures that the system can grow from a small pilot deployment to district-level and state-level operations without architectural redesign or disruption to ongoing operations.

7. Dashboards and User Interfaces

The system provides multiple dashboards tailored to different operational roles. Each dashboard is designed with a specific purpose and access scope, ensuring that users see only the information necessary for their responsibilities. Dashboards do not bypass backend controls; all access is mediated through authentication, role-based access control, and attribute-based access control mechanisms. The dashboard layer is a visualisation and interaction layer only. It does not perform analytics, store evidence, or make operational decisions.

Dashboards and Role Based User Interfaces



7.1 Edge Operator Dashboard

The edge operator dashboard is hosted on the operator laptop and is intended exclusively for drone operators. Its purpose is to support safe and effective drone operation rather than intelligence analysis. This dashboard provides real-time visibility into the live video feed, network connectivity, buffering status, and system health indicators such as frame rate and stream continuity. It also displays lightweight indicators

related to change detection and frame selection activity, enabling operators to confirm that the system is functioning correctly.

The edge dashboard does not display analytics results, alerts, or evidence. Operators cannot view or access confirmed incidents, forensic data, or analytics outputs. This restriction ensures that drone operators remain focused on flight safety and data capture and do not make operational judgements based on partial information.

7.2 Ingest and Streaming Operations Dashboard

The ingest and streaming operations dashboard is intended for system administrators and technical support personnel. It provides visibility into the health and performance of the ingest and streaming infrastructure. This dashboard displays information such as the number of active drone feeds, queue depths, throughput rates, latency indicators, and back-pressure conditions. It also shows service health metrics, error rates, and alerts related to infrastructure performance.

The operations dashboard does not provide access to live video, analytics outputs, or evidence clips. Its purpose is strictly operational monitoring and maintenance. This separation ensures that technical personnel cannot access sensitive policing data unless explicitly authorised through separate processes.

7.3 Central Command and Intelligence Dashboard

The central command and intelligence dashboard is the primary operational interface for control room officers, supervisors, and senior leadership. It consolidates confirmed analytics outputs, alerts, and evidence references from all active drones.

This dashboard provides real-time alerts, multi-drone situational views, crowd density visualisations, traffic analytics, and timelines of detected events. Authorised users can play evidence clips, review annotated frames, and access structured forensic logs directly from this interface. The dashboard supports filtering, search, and reporting functions to assist in operational review, investigation, and post-event analysis. All evidence playback and data access are streamed securely through backend APIs and are subject to access control and audit logging.

7.4 Role-Based Views and Access

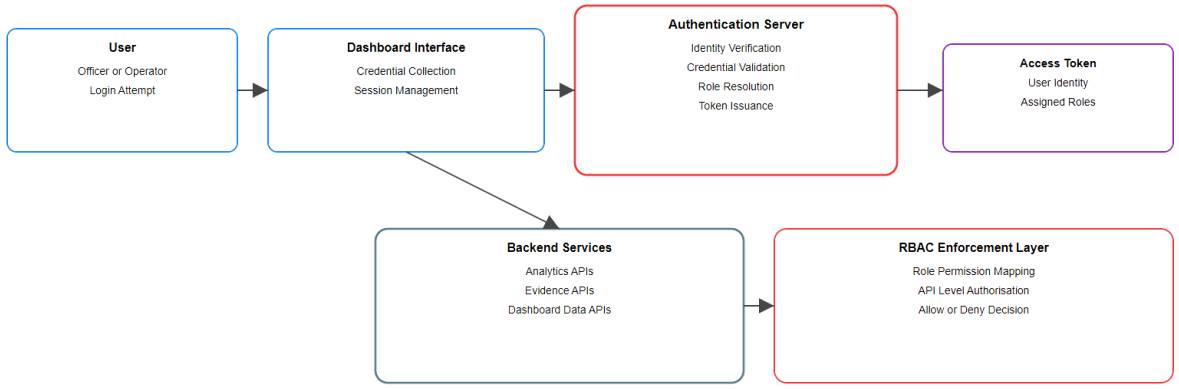
Access to dashboards and specific dashboard features is governed by role-based and attribute-based access control. Each user is assigned a role such as drone operator, control room officer, supervisor, senior officer, or system administrator. Roles define the baseline set of dashboard features available to the user. Attribute-based policies further refine access based on mission assignment, geographic jurisdiction, time window, and sensitivity of the data. For example, an officer may be permitted to view live alerts for an active mission but not access historical evidence outside their jurisdiction or time period. All dashboard interactions, including logins, views, playback actions, and data exports, are logged in the audit system. This ensures accountability, supports supervisory oversight, and aligns with legal and administrative requirements.

8. Authentication, Access Control and Security

Security and access control are foundational to the drone video analytics system. Given the sensitive nature of aerial surveillance data and digital evidence, the system enforces strict identity verification, contextual access control, and continuous auditability. Security is implemented as an integral part of the architecture rather than as an external or optional layer. The system follows a zero-trust approach in which every access request is authenticated, authorised, and logged, regardless of the user's role or location.

8.1 Authentication Mechanism

Authentication and Role Based Access Control Flow



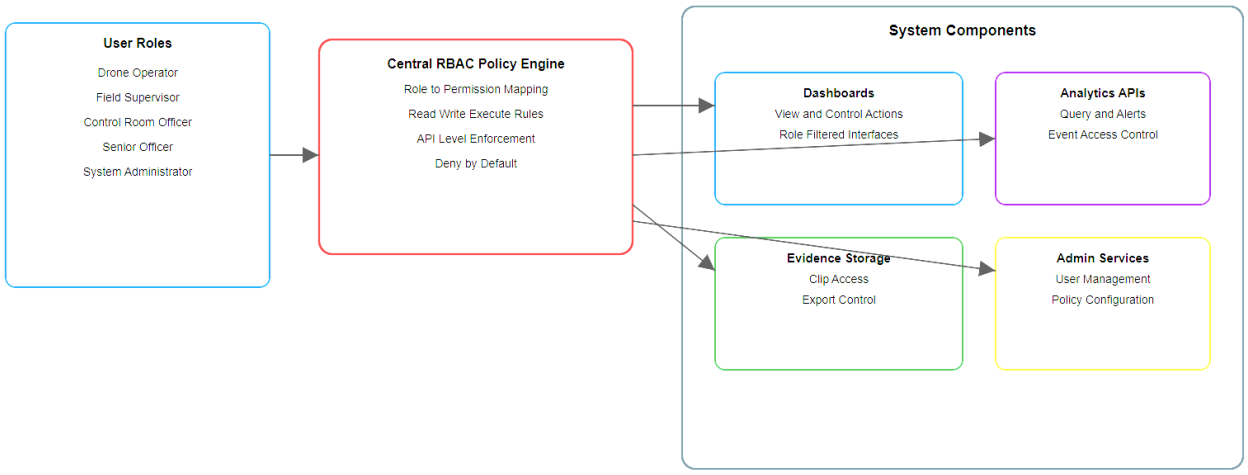
All users and system components accessing the platform are authenticated using standard, open authentication protocols. Authentication is handled through a central identity service supporting mechanisms such as OAuth 2.0 and OpenID Connect.

Users authenticate using individual credentials issued by the department. Devices such as operator laptops and ingest servers authenticate using system-issued credentials and certificates. Successful authentication results in the issuance of a short-lived security token that represents the authenticated identity. Authentication verifies *who* the user or system is but does not, by itself, determine what actions are permitted. Authorisation decisions are handled separately through access control mechanisms.

8.2 Role-Based Access Control (RBAC)

Role-Based Access Control defines baseline permissions based on the operational role assigned to each user. Typical roles include drone operator, control room officer, supervisor, senior officer, and system administrator.

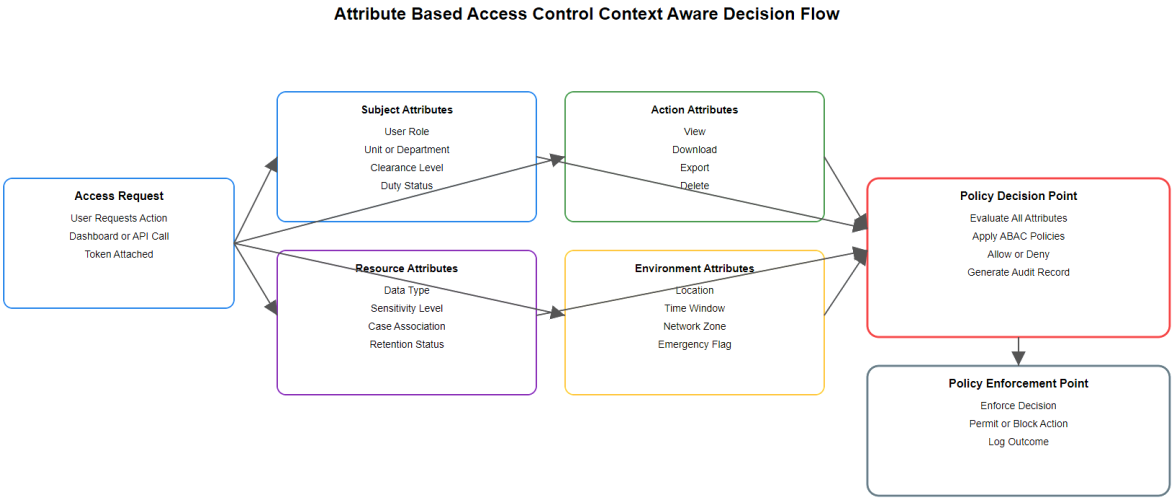
Role Based Access Control Enforcement Across System Components



RBAC determines which dashboards a user can access and which high-level actions are permitted. For example, drone operators can access the edge dashboard but cannot view analytics or evidence. Control room officers can view alerts and evidence related to active operations, while supervisors and senior officers have broader access for oversight and review. RBAC ensures that users cannot access system components or functions outside their official responsibilities, forming the first layer of authorisation control.

8.3 Attribute-Based Access Control (ABAC)

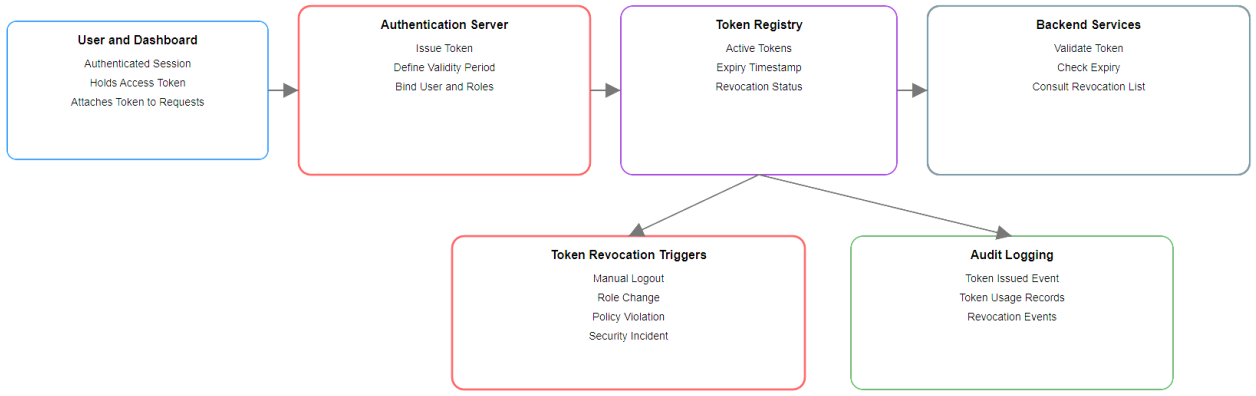
Attribute-Based Access Control provides fine-grained, contextual authorisation on top of RBAC. While RBAC answers *who the user is*, ABAC answers *whether access is appropriate in the current context*. ABAC policies evaluate multiple attributes at the time of each access request. These attributes include mission identifiers, geographic jurisdiction, time of access, sensitivity level of the data, and operational status of the event. For instance, an officer may have permission to view evidence related to an active mission within their jurisdiction but may be denied access to historical data from another district. ABAC ensures that access is dynamic, temporary, and context-aware. Even users with senior roles are restricted when contextual conditions are not met. This mechanism significantly reduces the risk of misuse and supports lawful, proportionate access.



8.4 Token Lifecycle and Revocation

After successful authentication, the system issues short-lived security tokens containing embedded claims such as role, authorised missions, and validity period. These tokens are attached to every API request and are validated by backend services before processing.

Token Lifecycle and Revocation Flow

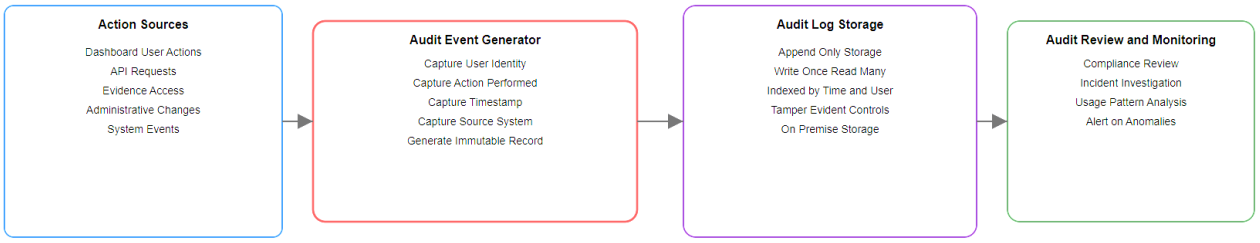


Tokens automatically expire after a defined duration, forcing periodic re-authentication. Tokens can also be revoked immediately in situations such as logout, device loss, role change, mission completion, or suspected compromise. Revocation mechanisms ensure that access can be terminated instantly across all system components. Backend services reject expired or revoked tokens, preventing further access without requiring system restarts or manual intervention.

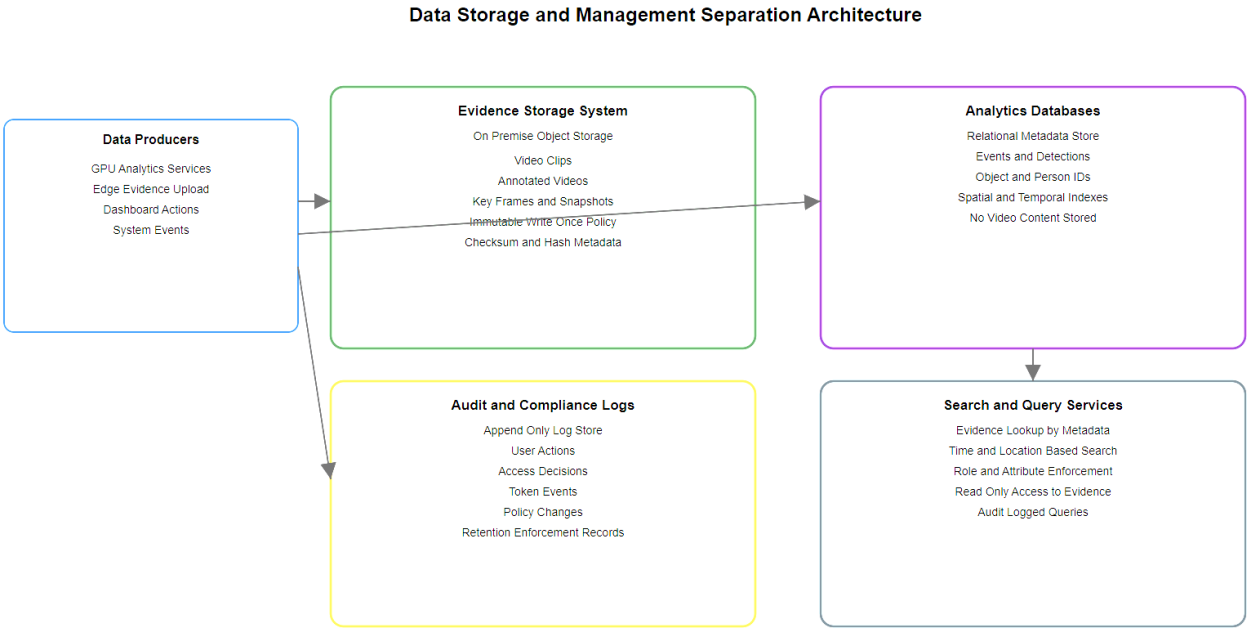
8.5 Audit Logging and Monitoring

Every significant system action is recorded in dedicated audit logs. This includes authentication attempts, dashboard access, analytics result views, evidence playback, evidence export, and administrative actions. Audit logs capture who performed the action, what data was accessed, when the action occurred, and from which system component. These logs are stored in tamper-evident systems and retained for extended periods as defined by policy. Audit monitoring supports supervisory oversight, internal reviews, and legal compliance. It ensures that all system usage is traceable and accountable, reinforcing trust in the system's operation and outputs.

Audit Logging and Monitoring Architecture



9. Data Storage and Management



The drone video analytics system follows a structured and policy-driven approach to data storage and management. Different categories of data are stored in different systems based on their purpose, sensitivity, access patterns, and legal requirements. This separation ensures performance efficiency, strong security, clear chain of custody, and compliance with data protection and evidence laws. The system explicitly avoids centralised storage of continuous raw video and retains only data that is operationally relevant and legally justified.

9.1 Evidence Video and Image Storage

Confirmed evidence video clips and annotated images are stored in a dedicated on-premise object storage system. This storage layer is designed for durability, integrity, and controlled access rather than high-throughput streaming.

Evidence clips are generated only after authoritative event confirmation by the GPU analytics cluster. Once created, each clip is uploaded to the object storage system in a standard video format and stored as an immutable object. Annotated frames and snapshots associated with the event are stored alongside the video clip to support quick review and visual verification. The object storage platform enforces retention policies and prevents modification or deletion of evidence outside authorised workflows. Access to stored evidence is mediated exclusively through backend APIs that apply authentication, role-based access control, and attribute-based access control. Direct access to storage systems is not permitted for users or dashboards.

9.2 Structured Analytics Databases

Structured outputs generated by the analytics engine are stored in relational databases. These outputs include detected events, entity records such as persons and vehicles, tracking identifiers, confidence scores, timestamps, and spatial metadata.

Relational databases are chosen for their strong consistency guarantees and support for complex queries required during investigation, reporting, and supervisory review. The database schema maintains clear relationships between events, source drones, evidence references, and operational contexts. Structured

analytics databases do not store raw video or image data. Instead, they store references to evidence objects stored in the object storage system, ensuring separation between metadata and media content.

9.3 Time-Series and Search Databases

Temporal analytics data, such as crowd density over time, traffic flow metrics, and system performance statistics, are stored in time-series databases. These databases are optimised for high-frequency writes and efficient aggregation over time windows.

Time-series data supports operational monitoring, trend analysis, and post-event review without overloading relational databases. Retention policies ensure that older time-series data can be aggregated or archived as required. Search databases are optionally used to index selected fields from structured databases to enable fast retrieval and filtering of events based on parameters such as location, time range, event type, or operational context. Search indexes improve user experience on dashboards without becoming a system of record.

9.4 Audit and Compliance Data Storage

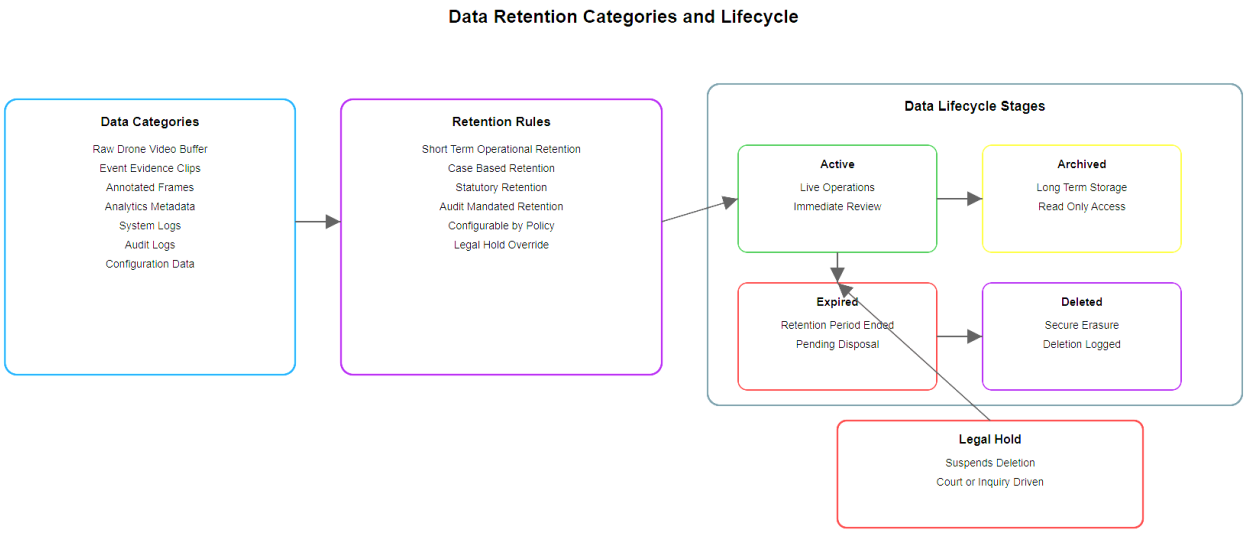
Audit and compliance data is stored in dedicated audit databases that are logically and, where feasible, physically separated from operational databases. Audit records capture authentication events, access attempts, evidence views, exports, administrative actions, and system decisions.

Audit data is treated as highly sensitive and is protected against tampering. Retention periods for audit logs are longer than those for operational data, reflecting legal and compliance requirements. Access to audit data is restricted to authorised supervisory and compliance roles. Audit logs support internal reviews, external audits, and court proceedings by providing a complete, traceable history of system usage and data access.

10. Data Retention Policy

The data retention policy defines how long different categories of data generated by the drone video analytics system are retained, where they are stored, and under what conditions they are archived or deleted. The policy is designed to balance operational requirements, legal obligations, privacy considerations, and infrastructure efficiency.

The system follows the principle of data minimisation. Continuous surveillance data is not retained, and only data that is operationally relevant and legally justified is stored beyond transient processing stages.



All data generated by the system follows a defined retention lifecycle based on its purpose and legal significance. Operational buffers are retained briefly, evidence clips are retained based on case requirements, analytics metadata is kept for trend and reporting needs, and audit logs follow statutory retention norms. Data transitions through active, archived, and expired stages, with secure deletion at the end of its lifecycle. At any point, legal hold can suspend deletion to comply with court orders or official inquiries, ensuring that no relevant data is lost.

10.1 Retention Categories

Data generated by the system is categorised based on its purpose and sensitivity. The primary categories include transient operational data, confirmed evidence data, structured analytics data, and audit and compliance data.

Transient operational data includes raw video streams and intermediate frames handled at the edge and ingest layers. This data is used only for immediate processing and is not considered evidence. Confirmed evidence data includes video clips and annotated images generated after authoritative event confirmation. This data forms part of the official record for investigation and review.

Structured analytics data includes metadata such as detected events, entity attributes, trajectories, timestamps, and confidence scores. This data supports operational analysis, reporting, and historical review. Audit and compliance data include authentication logs, access records, administrative actions, and system events required to demonstrate accountability and lawful usage.

10.2 Retention Durations and Rationale

Transient operational data, including continuous drone video and cognitively selected frames, is retained only for the minimum duration required for processing. Edge-level video buffers are overwritten continuously, typically within sixty seconds. Data held at the ingest layer is discarded shortly after analytics processing completes.

Confirmed evidence video clips and annotated images are retained for a configurable period, typically ranging from thirty to one hundred and eighty days. The retention duration may be extended for cases involving ongoing investigations, legal proceedings, or specific supervisory instructions.

Structured analytics data is retained for longer periods, often one to three years, as it supports trend analysis, operational review, and reporting without exposing raw video content. Time-series analytics data may have shorter retention periods, such as six to twelve months, with older data aggregated or archived.

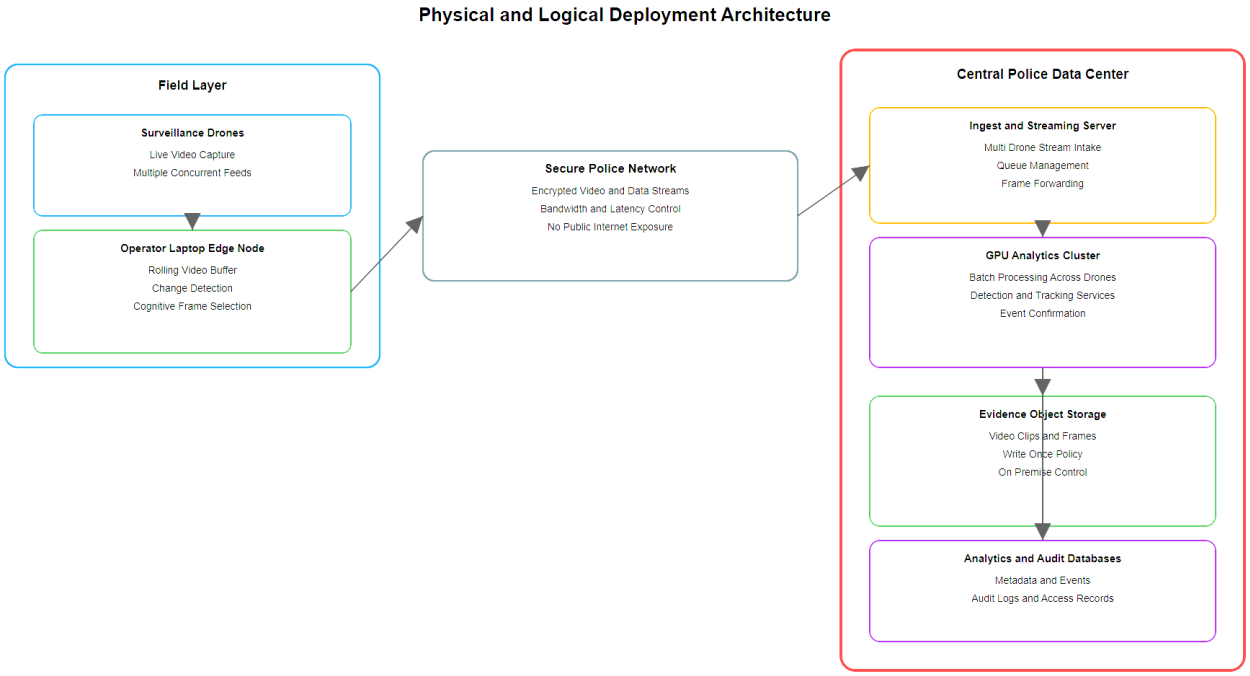
Audit and compliance data is retained for extended durations, typically a minimum of five years, reflecting legal, administrative, and accountability requirements. Longer retention may be applied where mandated by departmental policy or statutory obligations.

10.3 Configurability and Legal Hold

Retention policies are configurable at the system level and can be adjusted based on operational needs, legal requirements, or changes in policy. Configuration changes are restricted to authorised administrative roles and are recorded in audit logs.

The system supports legal hold mechanisms that override standard retention rules. When a legal hold is applied, affected evidence and associated metadata are exempt from automatic deletion until the hold is explicitly released. Legal holds may be applied in response to court orders, internal inquiries, or supervisory instructions. This approach ensures that data is neither retained unnecessarily nor deleted prematurely, maintaining compliance with privacy principles while preserving evidence when legally required.

11. Deployment Architecture



The deployment architecture defines how the drone video analytics system is physically and logically deployed within Kerala Police infrastructure. The architecture prioritises data sovereignty, operational control, security, and scalability by adopting a fully on-premise deployment model. All critical components are hosted within police-controlled environments and interconnected through secure internal networks. The deployment design supports phased expansion and ensures that operational, analytical, and evidentiary functions remain isolated where required, without compromising performance or manageability.

11.1 On-Premise Deployment Model

The system is deployed entirely on on-premise infrastructure owned or controlled by Kerala Police. No dependency on public cloud platforms is required for core operations, analytics, storage, or dashboards. This approach ensures full control over sensitive surveillance data, compliance with government data sovereignty expectations, and reduced exposure to external security risks.

Operator laptops function as edge nodes at the field level. Central infrastructure components such as ingest servers, GPU analytics clusters, evidence storage systems, databases, and dashboards are hosted at police headquarters or designated data centres. This centralisation of critical resources simplifies governance, security monitoring, and maintenance. Where required, private government networks or secure VPN links are used to connect distributed components, ensuring confidentiality and integrity of data in transit.

11.2 Separation of Ingest, Analytics and Storage

A key architectural principle in the deployment is the strict separation of ingest, analytics, and storage functions. Each of these functions is deployed as a distinct system component with independent scaling and maintenance characteristics.

The ingest and streaming servers are responsible solely for receiving, validating, and queuing data from edge devices. They are optimised for high-throughput, low-latency data handling and do not store evidence or perform analytics. The GPU analytics cluster is dedicated to computational processing and event confirmation. It consumes data from ingest queues, executes analytics models, and generates structured outputs. It does not retain video data or act as a storage system.

Evidence storage is implemented as a separate on-premise object storage platform optimised for durability, immutability, and controlled access. This separation preserves chain of custody and ensures that maintenance or scaling of ingest or analytics systems does not affect evidence availability. This clear separation reduces operational risk, simplifies audits, and allows each subsystem to evolve independently.

11.3 Network and Infrastructure Considerations

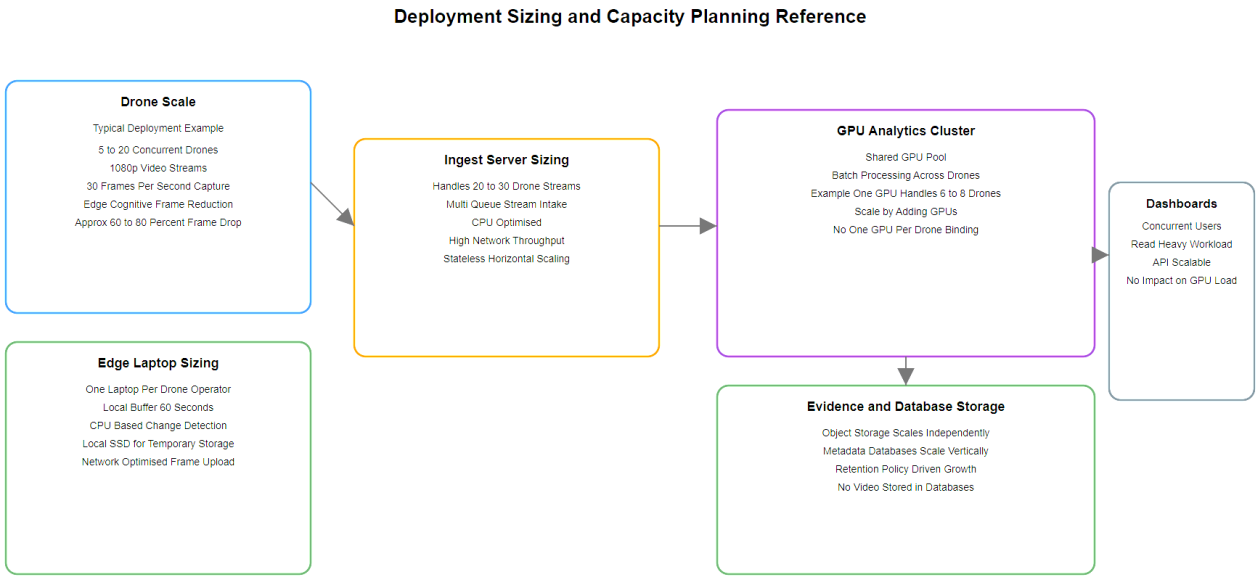
The deployment relies on a secure, high-bandwidth internal network connecting edge devices, ingest servers, analytics clusters, storage systems, and dashboards. Network segmentation is applied to isolate operational traffic, management traffic, and storage access.

Latency considerations are addressed by performing buffering and lightweight pre-processing at the edge and centralising GPU processing within a controlled environment. Bandwidth usage is optimised through cognitive frame selection, ensuring that network load remains manageable even with multiple concurrent drones.

Infrastructure components are provisioned with redundancy where feasible, including redundant network links, power supplies, and storage replication. Monitoring and alerting systems provide visibility into infrastructure health, enabling proactive maintenance and rapid response to failures. This deployment architecture ensures that the system remains resilient, secure, and capable of supporting sustained multi-drone operations under real-world conditions.

12. Deployment Sizing Example

This section provides an indicative sizing example for deploying the drone video analytics system in an on-premise environment. The objective is to demonstrate feasibility and planning logic rather than prescribe fixed hardware configurations. Actual sizing may vary based on operational intensity, terrain, camera specifications, and analytics configurations.



This deployment sizing example illustrates how system capacity scales across layers. Edge laptops reduce raw video load through buffering and cognitive frame selection, ensuring that only meaningful frames reach the central systems. The ingest server is designed to handle multiple concurrent drone streams and can be scaled horizontally. GPU resources are shared across drones using batch processing, avoiding one-to-one GPU allocation and optimising utilisation. Storage systems scale independently based on evidence retention policies, while dashboards remain lightweight and do not impact analytics throughput.

12.1 Assumptions

The sizing example assumes simultaneous operation of twenty drones deployed across different locations. Each drone streams high-definition video to an operator laptop over a stable wireless link. Operator laptops perform buffering, change detection, and cognitive frame selection but do not execute heavy analytics.

The system is assumed to operate during peak periods such as public events, traffic monitoring, or emergency response, with continuous drone operation for several hours. Analytics workloads include person detection, crowd estimation, vehicle tracking, and related features as described earlier. All infrastructure components are hosted on-premise at a central police facility, with secure internal networking and redundancy where feasible.

12.2 Edge Layer Sizing

Each drone is paired with one operator laptop. The laptop must be capable of decoding high-definition video, maintaining a rolling sixty-second buffer, executing lightweight pre-processing, and securely transmitting selected frames and evidence clips.

A typical configuration includes a modern multi-core CPU, at least sixteen gigabytes of RAM, solid-state storage for temporary buffering, and a dedicated graphics capability if available. The storage requirement at the edge is modest, as only short buffers and temporary evidence clips are stored before upload. Network connectivity must support stable upstream transmission of selected frames and periodic evidence clips. Since full video streams are not transmitted continuously to the backend, bandwidth requirements remain manageable.

12.3 Ingest Server Sizing

The ingest and streaming server must handle concurrent connections from all active operator laptops. It is optimised for network I/O, message queuing, and stream management rather than storage or compute-intensive tasks. A mid-range server with a multi-core CPU, sufficient memory for queue buffering, and high-speed network interfaces is typically adequate. Disk storage requirements are minimal, as the ingest server does not retain video or evidence data. For higher resilience, the ingest layer can be deployed with redundancy or as a small cluster, allowing failover and load balancing without impacting edge or analytics components.

12.4 GPU Cluster Sizing

The GPU analytics cluster is the most compute-intensive component of the system. It must support batch processing of frames from multiple drones while maintaining near real-time responsiveness. For an initial deployment with twenty drones, a small cluster with two to four high-performance GPUs is typically sufficient, depending on the models used and desired latency. GPUs are shared across all analytics services, and batch inference ensures efficient utilisation. The cluster is hosted on servers with adequate CPU resources, high memory capacity, and fast interconnects between GPUs. As drone count or analytics complexity increases, additional GPUs or nodes can be added without changes to upstream components.

12.5 Storage and Database Sizing

Evidence storage requirements depend on event frequency and retention policy rather than continuous recording. Since only confirmed events generate evidence clips, storage growth remains predictable and controlled. An object storage system with several tens of terabytes of usable capacity is typically sufficient for initial deployment, allowing for months of evidence retention with replication. Storage can be expanded incrementally by adding nodes or disks. Relational, time-series, and audit databases require comparatively modest storage but benefit from reliable hardware and regular backups. Database sizing should prioritise performance and integrity over raw capacity.

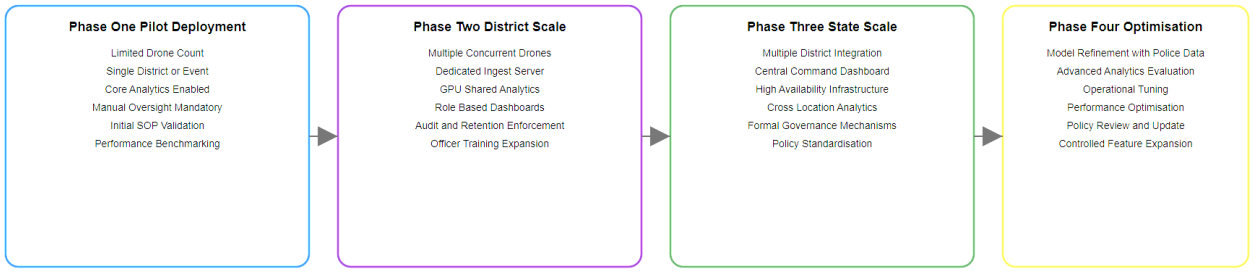
12.6 Dashboard and User Load

Dashboard usage is generally limited to a small number of concurrent users, including drone operators, control room officers, supervisors, and system administrators. The dashboard backend must support secure session management, data retrieval, and evidence streaming. A small set of application servers is typically sufficient to handle dashboard load, with capacity planned for peak usage during major events. Since dashboards consume processed data rather than raw video, their impact on overall system load is limited. This sizing approach ensures that the system can operate reliably under realistic operational conditions while remaining flexible for future expansion.

13. Phased Rollout Roadmap

The rollout of the drone video analytics system is planned in clearly defined phases to minimise operational risk, ensure user adoption, and allow incremental validation of technical and procedural elements. This phased approach enables Kerala Police to start with controlled deployments, gather operational feedback, and scale confidently without disrupting ongoing policing activities. Each phase has well-defined objectives, scope, and outcomes, ensuring transparency and accountability throughout the deployment lifecycle.

Phased Deployment Roadmap



The phased rollout roadmap shows how the system is introduced in a controlled and responsible manner. Initial pilot deployments validate core analytics, operational procedures, and legal compliance in a limited setting. District-level expansion introduces shared infrastructure, role-based dashboards, and audit enforcement. State-level deployment integrates multiple districts under a central command structure with high availability. The final phase focuses on optimisation and careful enhancement, ensuring that any advanced capabilities are introduced only after operational validation and policy approval.

13.1 Phase 1 – Pilot Deployment

The pilot deployment phase focuses on validating the core architecture, workflows, and analytics capabilities in real operational conditions. A limited number of drones, typically two to five, are deployed at selected locations such as a beach area, a sports venue, or a highway stretch. During this phase, operator laptops, ingest servers, GPU analytics, evidence storage, and dashboards are deployed in a controlled environment. Core analytics such as person detection, crowd density estimation, vehicle tracking, intrusion detection, and evidence generation are enabled. The objective of this phase is to validate system stability, latency, accuracy, and usability. Feedback from drone operators, control room officers, and supervisors is collected to refine workflows, dashboards, and operational procedures. Legal and audit processes, including evidence handling and access logging, are also verified during this phase.

13.2 Phase 2 – District-Level Deployment

The district-level deployment phase expands the system to support multiple drones operating concurrently within a district. Additional operator laptops and ingest capacity are provisioned, and GPU resources are scaled as required. During this phase, the system is integrated more closely with district-level command and control operations. Dashboards are used regularly by control room officers for monitoring live operations and reviewing evidence. Operational standard operating procedures (SOPs) are formalised based on pilot experience. Training programs are conducted for operators and supervisory staff. The focus remains on reliability, consistency, and procedural compliance rather than feature expansion.

13.3 Phase 3 – State-Level Deployment

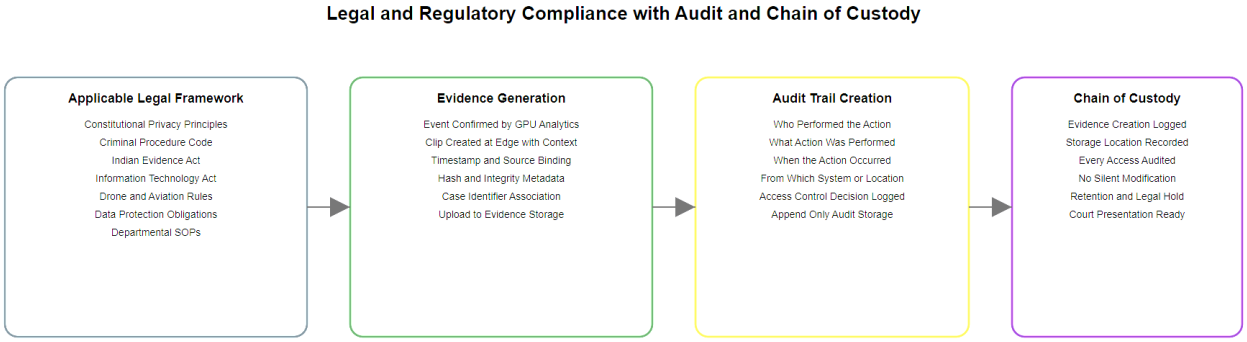
The state-level deployment phase extends the system across multiple districts, supporting a larger fleet of drones operating simultaneously from different locations. Central infrastructure at headquarters is scaled to handle increased ingest, analytics, and storage demands. This phase introduces coordinated monitoring and reporting at the state command level. Senior officers gain access to aggregated dashboards that provide situational awareness across districts while maintaining jurisdictional and contextual access controls. Inter-district analytics consistency, audit processes, and data governance mechanisms are reinforced. This phase establishes the system as a core operational capability rather than a pilot or specialised tool.

13.4 Phase 4 – Optimisation and Advanced Capabilities

The final phase focuses on optimisation, refinement, and selective introduction of advanced analytics capabilities. Performance tuning, model calibration using Kerala Police–specific data, and workflow optimisation are prioritised. Advanced capabilities such as improved crowd behaviour analysis in drone video, enhanced cross-camera association, and altitude-adaptive models may be introduced based on operational need and data readiness. These enhancements are added without disrupting core functionality. Continuous monitoring, periodic audits, and regular training updates ensure that the system remains effective, compliant, and aligned with evolving operational requirements.

14. Legal and Regulatory Compliance

The drone video analytics system has been designed with explicit consideration of Indian constitutional principles, statutory requirements, and regulatory frameworks governing surveillance, evidence handling, data protection, and aviation. Legal compliance is treated as a foundational requirement rather than a post-deployment adjustment.



The system architecture, workflows, access controls, and data retention mechanisms collectively ensure that technology use remains lawful, proportionate, and accountable.

14.1 Constitutional and Privacy Compliance

The system aligns with constitutional protections under Articles 14, 19, and 21 of the Constitution of India, particularly the right to privacy as recognised by the Supreme Court. Surveillance is conducted only for legitimate policing purposes such as public safety, traffic management, and emergency response. The design follows principles of necessity and proportionality. Continuous surveillance footage is not retained, and analytics focus on situational awareness rather than personal profiling. Access to analytics outputs and evidence is strictly controlled, logged, and limited to authorised personnel. By minimising data collection and restricting access through technical and procedural safeguards, the system supports lawful use while reducing the risk of arbitrary or excessive surveillance.

14.2 Criminal Procedure and Evidence Law

Evidence generated by the system is handled in a manner consistent with the Code of Criminal Procedure and applicable evidentiary standards. Video clips and annotated images are created only after authoritative confirmation by the analytics system and are stored in tamper-resistant object storage. Chain of custody is preserved through controlled access, immutable storage, and comprehensive audit logging. Each evidence item is traceable to its source drone, time of capture, and authorised creation event. For use in legal proceedings, evidence can be accompanied by appropriate certification, including compliance with Section 65B of the Indian Evidence Act where electronic records are relied upon. The system's audit logs and metadata support authenticity and integrity verification.

14.3 Information Technology and Cyber Laws

The system complies with applicable provisions of the Information Technology Act and related rules governing electronic records, data security, and access controls. Secure authentication mechanisms, encrypted communication, and role-based access ensure that unauthorised access is prevented. System logs and audit trails support accountability and incident investigation in the event of misuse or breach. Cybersecurity practices such as network segmentation, access monitoring, and regular updates further strengthen compliance.

14.4 Data Protection Regulations

Data handling within the system follows principles consistent with emerging data protection frameworks in India. Data minimisation, purpose limitation, and retention control are embedded into the design. Personal data is processed only where operationally necessary and is not retained beyond defined retention periods unless legally required. Evidence and analytics data are stored within police-controlled infrastructure, ensuring data sovereignty and reducing exposure to third-party risks. The system supports legal holds and supervisory oversight, enabling compliance with lawful data preservation requests while avoiding unnecessary long-term retention.

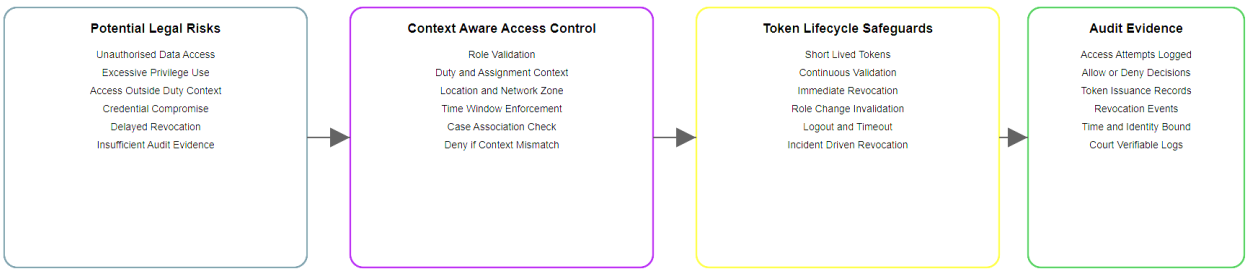
14.5 Drone and Aviation Regulations

Drone operations conducted as part of the system comply with applicable civil aviation and unmanned aircraft regulations issued by competent authorities. The system does not interfere with drone flight control or navigation and does not impose additional processing loads on the drone hardware. By restricting processing and storage to ground-based infrastructure, the system supports compliance with operational and safety guidelines applicable to drone deployment by law enforcement agencies.

15. Legal Risk Mitigation Strategy

The drone video analytics system incorporates legal risk mitigation as a core design objective. Rather than addressing risks reactively, the system embeds technical, procedural, and governance safeguards at every stage of the data lifecycle. This approach ensures that operational benefits are achieved without exposing Kerala Police to avoidable legal, constitutional, or reputational risks. The risk mitigation strategy addresses surveillance and privacy concerns, evidence integrity, cybersecurity threats, and operational accountability.

Legal Risk Mitigation through Access Control and Token Safeguards



This diagram shows how legal risks arising from misuse or overreach are mitigated through layered technical safeguards. Access is not granted based on role alone; it is continuously evaluated against contextual attributes such as duty status, location, and case association. Even when access is granted, it remains temporary through short-lived tokens that are validated on every request. Tokens can be revoked immediately in response to role changes, policy violations, or security incidents. Every access decision

and revocation event is logged in an immutable audit system, ensuring that the organisation can demonstrate due diligence, proportionality, and accountability at any point in time.

15.1 Surveillance and Privacy Risks

Uncontrolled or excessive surveillance presents significant legal and constitutional risks. To mitigate these risks, the system avoids continuous retention of raw drone video and limits permanent storage to confirmed, operationally relevant events. Analytics outputs are contextual and event-based rather than individual-centric. The system does not perform mass identification or persistent personal profiling. Face detection and person tracking are used as analytical aids and are subject to visibility, confidence, and legal authorisation constraints. Access to surveillance outputs is tightly controlled through RBAC and ABAC, ensuring that only authorised officers with a legitimate operational role can view data. Every access is logged and reviewable, discouraging misuse and supporting accountability. These safeguards collectively ensure that surveillance remains proportionate, purpose-bound, and defensible under constitutional scrutiny.

15.2 Evidence Integrity Risks

Digital evidence is vulnerable to challenges related to authenticity, tampering, and chain of custody. The system mitigates these risks by generating evidence only after authoritative event confirmation and storing it in immutable, on-premise object storage. Evidence is never modified after creation. All interactions with evidence, including viewing, playback, export, and deletion under policy, are logged with user identity, time, and purpose. Structured metadata links each evidence item to its source drone, timestamps, and system decisions. These controls ensure that evidence integrity can be demonstrated clearly during internal reviews or court proceedings, reducing the risk of exclusion or challenge.

15.3 Cybersecurity Risks

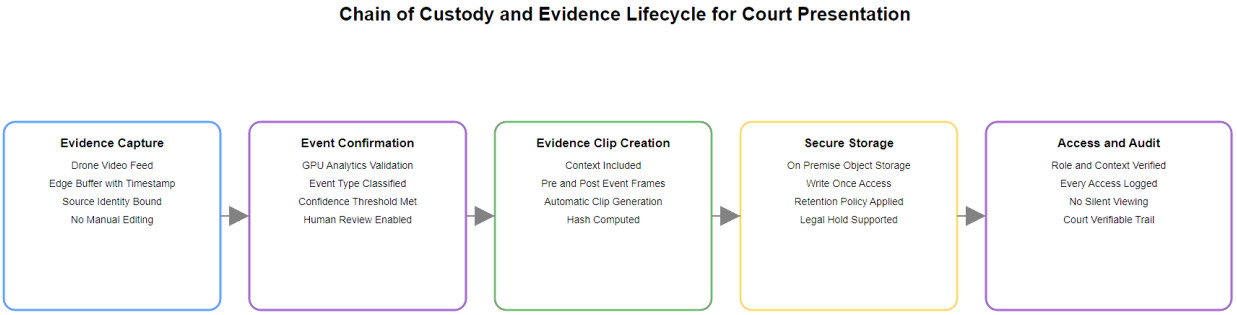
Cybersecurity risks include unauthorised access, data leakage, and system compromise. The system mitigates these risks through a zero-trust security model, strong authentication, encrypted communication, and network segmentation. No component trusts another by default. All requests are authenticated and authorised, including machine-to-machine communication. Short-lived tokens and revocation mechanisms limit exposure in case of credential compromise. Audit logs and monitoring systems enable early detection of anomalies and support incident response. Regular updates, vulnerability management, and restricted administrative access further reduce cybersecurity risk.

15.4 Operational and Audit Risks

Operational risks arise when systems are used inconsistently, without clear procedures, or without adequate oversight. The system mitigates these risks by defining clear roles, workflows, and dashboards aligned with operational responsibilities. Separation of duties ensures that no single role controls capture, analysis, storage, and access simultaneously. Supervisory oversight is enabled through dashboards and audit trails. Audit logs provide a complete record of system usage, supporting internal audits, external inspections, and judicial review. This transparency strengthens institutional trust and reduces the risk of adverse findings during audits or inquiries.

16. Chain of Custody and Court Presentation

The drone video analytics system has been designed to ensure that digital evidence generated through drone operations meets the requirements of authenticity, integrity, and traceability expected under Indian law. Chain of custody is preserved through a combination of technical controls, procedural safeguards, and audit mechanisms, ensuring that evidence can be confidently relied upon during judicial proceedings.



16.1 Evidence Creation and Preservation

Evidence is created only after a confirmed event is generated by the central GPU analytics cluster. The confirmation is based on defined confidence thresholds and consistency checks, ensuring that evidence generation is not triggered by transient or ambiguous signals. Once confirmation is received, the operator laptop extracts a continuous video segment from its rolling buffer, including a defined pre-event and post-event duration. This clip represents a faithful and uninterrupted record of the incident context. The clip is immediately uploaded to the central on-premise evidence storage system. Upon upload, the evidence object is stored in immutable form. No editing, overwriting, or replacement is permitted. The original file checksum, creation timestamp, source drone identifier, and system-generated metadata are recorded and preserved. This process ensures that the evidence remains unchanged from the moment of creation.

16.2 Access and Audit Trail

Access to evidence is strictly controlled through backend APIs governed by authentication, role-based access control, and attribute-based access control. Evidence cannot be accessed directly through storage systems or file paths. Every interaction with evidence, including viewing, playback, download, or export, is logged in the audit system. Audit records capture the identity of the user, role, time of access, purpose, and the specific evidence item involved. Audit logs are tamper-evident and retained for extended durations. Supervisory officers and authorised compliance personnel can review these logs to verify proper handling and to investigate any irregularities.

16.3 Court-Friendly Chain of Custody Explanation

For court presentation, the chain of custody can be explained clearly as follows. The video was captured by a police-operated drone during lawful duty. The footage was temporarily buffered and processed by a police-controlled system. Upon automated confirmation of a relevant incident, a short video clip was created automatically and stored in a secure police evidence repository.

From that point onwards, the video remained unchanged. Every access to the video was recorded automatically by the system, and only authorised officers were permitted to view or retrieve it. The system maintains complete records showing when the video was created, where it was stored, and who accessed it.
This explanation establishes continuity, authenticity, and control, satisfying evidentiary expectations under the Indian Evidence Act, including requirements for electronic records.

17. Training and SOP Compliance

Effective use of the drone video analytics system depends not only on technology but also on proper training and adherence to standard operating procedures. The system is accompanied by structured training programs and clearly defined SOPs to ensure consistent, lawful, and accountable operation.

17.1 Training Requirements by Role

Drone operators receive training focused on safe drone operation, system connectivity, buffer awareness, and handling of edge dashboards. They are explicitly trained not to interpret analytics outputs or access evidence beyond their role. Control room officers are trained in interpreting analytics alerts, reviewing evidence clips, and using dashboards for situational awareness. Training emphasises the advisory nature of analytics outputs and the importance of corroboration with ground inputs. Supervisors and senior officers receive training on oversight functions, evidence review, audit log interpretation, and legal considerations. System administrators receive training on infrastructure management, security controls, and incident response procedures.

17.2 Standard Operating Procedures

Standard operating procedures define how drones are deployed, how video analytics are used, how evidence is generated and reviewed, and how incidents are escalated. SOPs clearly specify roles, responsibilities, and approval chains. Procedures also define actions to be taken in exceptional situations, such as system outages, suspected misuse, or legal holds. SOP adherence is mandatory and forms part of routine operational review.

17.3 Compliance and Disciplinary Measures

Compliance with SOPs and system usage policies is enforced through technical controls and audit monitoring. Deviations from authorised workflows are detectable through audit logs and system alerts. Violations, whether technical or procedural, are subject to internal review and disciplinary action in accordance with departmental rules. This enforcement framework reinforces accountability and ensures that the system is used responsibly and lawfully.

Conclusion and Way Forward

The drone video analytics system described in this document represents a carefully balanced integration of technology, operations, and legal safeguards tailored for real-world policing needs. The design moves beyond a technology demonstration and establishes a structured operational intelligence platform that supports Kerala Police in managing complex, large-scale, and time-sensitive situations.

By combining drone-based sensing, edge buffering and pre-processing, centralised ingest and GPU-based analytics, secure evidence storage, and role-based dashboards, the system ensures that actionable insights are delivered without compromising data integrity, privacy, or accountability. Each architectural and workflow decision has been guided by principles of proportionality, separation of responsibilities, data minimisation, and human-in-the-loop decision-making.

A key strength of the system is its scalability. The architecture supports multiple drones operating simultaneously across different locations, efficient GPU utilisation through batch processing, and independent scaling of ingest, analytics, storage, and dashboards. This ensures that the system can grow from pilot deployments to district-wide and state-wide operations without fundamental redesign.

Equally important is the system’s legal and institutional robustness. Chain of custody is preserved end-to-end, access is tightly controlled and auditable, and retention policies align with constitutional principles, criminal procedure, and data protection expectations. The system is designed to withstand judicial scrutiny, internal audits, and public accountability requirements.

The phased rollout roadmap provides a practical path for adoption, allowing Kerala Police to validate performance, refine SOPs, and build institutional confidence before large-scale deployment. Training and compliance frameworks ensure that personnel across roles understand not only how to use the system, but also the legal and ethical boundaries governing its use.

Going forward, the immediate next steps involve initiating pilot deployments in selected scenarios, formalising SOPs based on operational feedback, and establishing governance mechanisms for oversight and review. As confidence and data maturity increase, the system can be optimised further and selectively enhanced with advanced analytics, always subject to legal review and operational necessity.

In conclusion, this system positions Kerala Police to harness drone-based video analytics as a responsible, scalable, and legally sound capability. It strengthens situational awareness, improves response effectiveness, and enhances evidentiary reliability, while ensuring that authority, judgement, and accountability remain firmly with trained officers and established legal processes.

Annexure A

Training and Standard Operating Procedure (SOP) Compliance Note for Officers

This system is designed to support policing operations only when used in accordance with approved training and standard operating procedures. To ensure lawful, effective, and consistent use, all officers interacting with the system must undergo role-appropriate training before operational access is granted.

Training Requirements

Drone operators shall be trained in safe drone operation, lawful use of aerial surveillance, system startup and shutdown procedures, and interpretation of edge-level indicators. Operators must clearly understand that they do not make intelligence or enforcement decisions based on the system and that their role is limited to safe operation and data transmission.

Control room officers shall be trained in dashboard usage, interpretation of alerts, analytics outputs, and evidence playback. Training shall emphasise that system outputs are decision-support tools and must be interpreted in conjunction with ground realities and supervisory guidance.

Supervisory and senior officers shall receive training on access controls, audit review, evidence handling protocols, and oversight responsibilities. This includes understanding when access to sensitive data is justified and how to review audit logs for compliance.

System administrators shall be trained in infrastructure management, cybersecurity hygiene, backup procedures, and incident response. Administrators shall not access operational evidence unless explicitly authorised for audit or maintenance purposes.

SOP Compliance

Officers shall access the system only using their assigned credentials and authorised devices. Sharing of credentials or unauthorised access is strictly prohibited. Officers must ensure that access is limited to the scope of their assigned duty, mission, and jurisdiction.

Evidence clips generated by the system shall not be downloaded, copied, or shared outside authorised channels. Any export of evidence shall follow written approval procedures and shall be logged automatically by the system.

Officers shall not attempt to alter, edit, or delete any evidence stored in the system. Any suspected system malfunction, false alert, or data anomaly must be reported immediately to the designated supervisory authority.

Periodic refresher training and compliance reviews shall be conducted to ensure continued adherence to SOPs. Failure to comply with training or SOP requirements may result in suspension of system access and disciplinary action as per departmental rules.

Annexure B

Model Certificate under Section 65B of the Indian Evidence Act, 1872
(To be adapted on official letterhead as required)

Certificate under Section 65B of the Indian Evidence Act, 1872

I, _____, son/daughter of _____, aged _____ years, presently working as _____ in the _____ (designation and unit), do hereby certify as follows:

1. That I am in charge of and responsible for the operation and maintenance of the drone video analytics system used by the Kerala Police, and I am competent to issue this certificate.
2. That the electronic record produced herein, namely the video clip(s) and/or digital data pertaining to _____ (brief description of incident), was produced by a computer system during the ordinary course of its operation.
3. That the said computer system was operating properly at the time of creation of the electronic record, and to the best of my knowledge and belief, there was no such malfunction which would affect the accuracy of the contents of the electronic record.
4. That the electronic record was generated automatically by the system from data received from authorised drone devices and processed through designated system components without manual alteration.
5. That the information contained in the electronic record is derived from information fed into the computer system in the ordinary course of lawful policing activities.
6. That the electronic record has been stored in a secure, access-controlled system, and its integrity has been preserved through audit logging and access controls.
7. That the copy of the electronic record produced is a true and accurate representation of the original electronic record stored in the system.

I hereby certify that the above statements are true to the best of my knowledge and belief.

Place: _____
Date: _____
Signature: _____
Name: _____
Designation: _____

Seal of Office

Section–Figure Mapping Table

Section No.	Section Title	Figure No(s).	Figure Title / Purpose
2	System Overview	Fig. 1	Overall Integrated Drone Video Analytics System Architecture
3	End-to-End System Architecture	Fig. 1	Shows complete system components and their relationships
3.3	Operator Laptop (Edge Layer)	Fig. 3	Edge processing, buffering, change detection, cognitive frame selection
3.4	Central Ingest and Streaming Server	Fig. 4	Ingest server architecture and stream management
3.5	GPU Analytics Cluster	Fig. 5	GPU-based multi-drone batch analytics
3.6	Evidence Storage System	Fig. 6	On-premise object storage for evidence clips and frames
3.7	Databases and Audit Systems	Fig. 7	Structured databases and audit logging architecture
3.8	Central Command Dashboard	Fig. 8	Dashboard layers, roles, and access separation
4	Operational Workflow	Fig. 2	End-to-end operational flow from drone to dashboard
4.2	Change Detection and Cognitive Frame Selection	Fig. 3	Logical separation and interaction of edge processes
4.4	GPU-Based Analytics and Event Confirmation	Fig. 5	How analytics confirm events centrally
4.5	Evidence Clip Creation and Upload	Fig. 12	Evidence generation and storage pipeline
5	Analytics Capabilities	Fig. 5	Mapping analytics features to GPU processing
6	Multi-Drone Processing and Scalability	Fig. 5	Batch processing and scheduling across drones
7	Dashboards and User Interfaces	Fig. 8	Different dashboards and role-based access
8.1	Authentication Mechanism	Fig. 9	Authentication and RBAC flow
8.2	Role-Based Access Control (RBAC)	Fig. 9	Role enforcement across system components
8.3	Attribute-Based Access Control (ABAC)	Fig. 10	Context-aware access decision flow
8.4	Token Lifecycle and Revocation	Fig. 11	Token issuance, usage, expiry, and revocation
8.5	Audit Logging and Monitoring	Fig. 7	Audit trail generation and storage
9	Data Storage and Management	Fig. 6, Fig. 7	Separation of evidence storage and databases
10	Data Retention Policy	Fig. 13	Retention categories and lifecycle
11	Deployment Architecture	Fig. 1	Physical and logical deployment layout
12	Deployment Sizing Example	Fig. 1, Fig. 5	Capacity planning reference diagrams
13	Phased Rollout Roadmap	Fig. 14	Phased deployment progression
14	Legal and Regulatory Compliance	Fig. 7, Fig. 15	Audit and chain-of-custody relevance
15	Legal Risk Mitigation Strategy	Fig. 10, Fig. 11	Access control and revocation safeguards
16	Chain of Custody and Court Presentation	Fig. 15	Evidence lifecycle and integrity explanation